

Enterprise Correspondence Management - Letters -- Integration Architecture

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Solution Metadata

Name	Value
Solution Status	DRAFT
Solution Type	NEW TENANT
Tenant	TBD ECM-L
Recommended Namespace	ecml
Recommended BIP Tenant App(s)	bip-ecml-resource-manager bip-ecml-resource-manager-mfe-ui bip-ecml-letter-manager bip-ecml-letter-manager-mfe-ui bip-ecml-template-manager bip-ecml-template-manager-mfe-ui bip-ecml-svc bip-ecml-mfe-root-ui
Sponsoring organization	OIT
Number of users	TBD
Estimated Monthly Cost	TBD

Privacy	<i>Differs from system to system.</i> <i>Resource Manager - None.</i> <i>Template Manager - None.</i> <i>Letter Manager - Both possible.</i> <i>ECM-L-API - Both possible.</i>
Will need 508 Compliance	<i>Yes, all of the MFE UI's will require 508 compliance.</i>
Deployment date	<i>TBD</i>

Introduction

This document outlines the design for Enterprise Correspondence Management-Letters (ECM-L), a centralized platform that empowers business users to manage the content and generation of letters efficiently. Currently, multiple Lines of Business (LoBs) within the Veterans Affairs infrastructure use disparate methods and applications to create correspondence, leading to operational inefficiencies and a heavy dependence on IT support.

ECM-L addresses these challenges by providing a unified solution that enables LoBs to self-manage their correspondence needs within a standardized environment. Once onboarded to ECM-L, LoBs can independently make changes—such as updating Letter Templates and Resources—without requiring IT intervention. This centralized approach ensures a controlled, reliable environment for producing, testing, releasing, and storing correspondence, streamlining operations and enhancing consistency across all LoBs.

What Problem Needs to be Solved?

Individual Lines of Business (LoBs) within the Veterans Affairs Organization utilize disparate methods and applications to generate their own correspondence. Without a centralized Enterprise Correspondence Management system, each LoB operates independently for Letter Templates and changes, and any substantial changes to Templates or mandates rely on IT resources for those changes, often requiring customized development for each LoB. This fragmented approach creates resource bottlenecks, operational inefficiencies, and limited historical tracking, ultimately resulting in delays and potential inaccuracies in the correspondence provided to our Veterans.

Business Needs

Is this a defect or an enhancement?	Enhancement
Who reported the need?	Office of Information & Technology (OIT)
When was the need noticed?	Q2 2022
When did/will it become a problem?	Q2 2024
Is there an existing Requirement? Strategic Objective?	TBD
What existing resources, tickets, emails, etc. might serve to better understand the problem?	
What business processes are involved?	Document Generation, Document Storage and Retrieval
Who is the business owner for the impacted systems or processes?	• Jeffry Boutet • Brian Steege • Peter Graziatis
Who are the business decision makers for solving the problem?	Pam Devine
What is the historical context? How did we get where we are?	Historically, VA has used multiple applications to generate, manage, and edit letters. Correspondence Service has provided document generation functionality to the VBMS system, providing document generation for use in VBMS Core, Ratings, and Awards. With this architecture, any changes needed by the business specific letter requires large amounts of commitment and OIT resources.

What happens to the business stakeholders if nothing is done?	VA will have disjointed efforts and applications for correspondence and letters
Of the various aspects of this problem, which of them does the business consider as critical flaws? Which are nice-to-have changes?	• Need for improvements in Letters Finalization • Need for centralized enterprise system that allows multiple LoBs to draft, generate, and Finalize Letters

Costing Considerations

Development Costs

- **Initial development:** \$8.4M
 - Staff costs: \$7.2M
 - Tools and licenses: \$0.7M
 - Training: \$0.3M
 - Contingency: \$0.2M

Infrastructure Costs

- **AWS GovCloud infrastructure:** \$0.9M Per Year
 - Compute: \$0.4M
 - Storage: \$0.3M
 - Database: \$0.1M
 - Networking/Other: \$0.1M

Maintenance Costs

- **Annual maintenance:** \$2.1M Per Year
 - Staff costs: \$1.2M
 - Infrastructure updates: \$0.3M
 - License renewals: \$0.1M
 - Ongoing enhancements: \$0.5M

5-year Total Cost of Ownership

- Development: \$8.4M
- Infrastructure: \$4.5M (\$0.9M Per Year x 5 years)
- Maintenance: \$9.5M (\$2.1M Per Year x 4.5 years)
- **Total: \$22.4M**

Chargeback Considerations

Each Line of Business (LoB), independent of one another, requires their own Resources, Templates, and Letter Instances in ECM-L; therefore, they can be easily separated financially by an LoB's application.

Technical Needs

Historically, VA has used multiple applications to generate, manage, and edit letters. This causes numerous workflow inconsistencies, various vendors, and large amounts of OIT commitments and resources, none which seem to have a perfected process in place. The purpose of Enterprise Correspondence Management- Letters (ECM-L) is to create a centralized, enterprise system, built around shared concepts, processes, and workflows to be leveraged by multiple Lines of Business (LoBs) in the Veterans Benefits Administration (VBA). The system will be LoB oriented, each being responsible for their integration and usage, with BID only supporting, facilitating, and trouble-shooting as needed. Onboarding new LoB will not require development work, but use configurations and system activities.

In order to prevent unintended interactions or dependencies, as well as support charge-back capabilities, the content, processes, and data of each LoB will be completely separate from those of another LoB. In addition, the ECM-L will be LoB agnostic, providing reasonable common denominators in concepts or processes among the LoBs, but ECM-L will not contain any proprietary business logic or concepts. Each LoB will be able to integrate and incorporate their own business logic or processes through predefined hooks, APIs, or other integration points. This will allow API access to Letter Instances in data format as well as give LoBs the ability to configure data lookup "callback" to its own systems.

The implications of this enterprise system allow for reusing existing concepts in its approach, the opportunity to consolidate and simplify letter generation and management across VBA, and the ability to plan for complexity.

Describe the problem in technical terms	We need an Enterprise level Line of Business agnostic tool to create and manage correspondence that can integrate with external API's without the need for a code change.
What are the architectural components in the problem's context?	New tenants: Resource Manager, Template Manager, Letter Manager, ECM-L Service.
What are the integration points that might be impacted by this problem?	Each of the new tenants will integrate with Line of Business specific API's on an as needed basis.
What technologies are involved?	Springboot, React, Kafka, iText, Lexical.
Is this related to already-stated technical debt?	No.
If the problem is with a technology, what were the reasons for using this technology in the first place?	N/A
Has an attempt been made before to fix the problem?	No. The existing components in the DocGen namespace were not designed to accomplish this goal / solve this problem and are too entwined with Line of Business specific logic to be repurposed to solve it.
Did it fail or did it just not finish?	N/A.
Who are the technical PoCs for that attempt?	N/A.
What components are impacted by this problem?	-All VA letters/correspondence -Core, Ratings, Awards
Are there current mitigations that help alleviate the impact of the problem?	No.
What are the technical dependencies for this? What does this depend on? What depends on it?	The new tenants proposed will stand on their own and external Line of Business specific API's can be configured in as dependencies as needed.
Based on the SV-1 and SV-9 generated what is estimated monthly cost for this new service? Upload estimate of resources from the AWS Pricing Calculator	TBD

Solution Space and Discussion

The solution space describes the architectural context related to the problem/solution in a 4+1 style document. A 4+1 Design is a human-readable, standardized way of viewing an architecture from multiple perspectives. Each perspective is complementary, offering a different way to look at a particular architecture. Taken as a whole, a 4+1 Design is a comprehensive way to communicate an architecture.

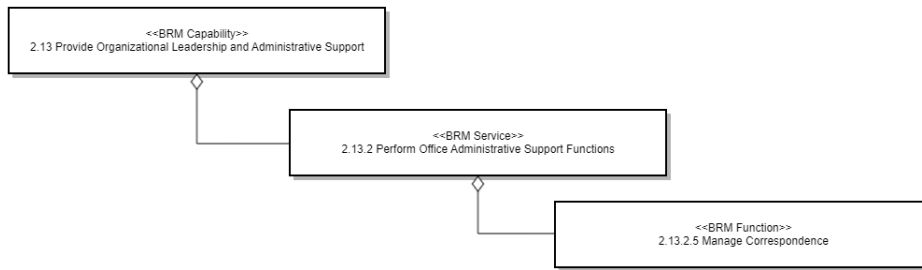
Use Case View

Capability Viewpoint Vision (CV-1) (VASI Description)

The Enterprise Correspondence Management - Letters (ECM-L) service will reside on the Benefits Integration Platform (#2295) and under the Veterans Benefits Management System (VBMS #1728) Application. ECM-L will deliver an enterprise-level document management system accessible to multiple Lines of Business within the Veterans Benefits Administration (VBA), focusing on enterprise design principles. The ECM-L components will allow for each Line of Business to independently manage Templates, create drafts of, and finalize correspondence. ECM-L is highly extensible, enabling LoBs to integrate its ecosystem into their own business logic through predefined hooks, API's, and other integration points. Ultimately, ECM-L provides a unified and Line of Business agnostic method for creating letters, streamlining correspondence generation across all VBA systems.

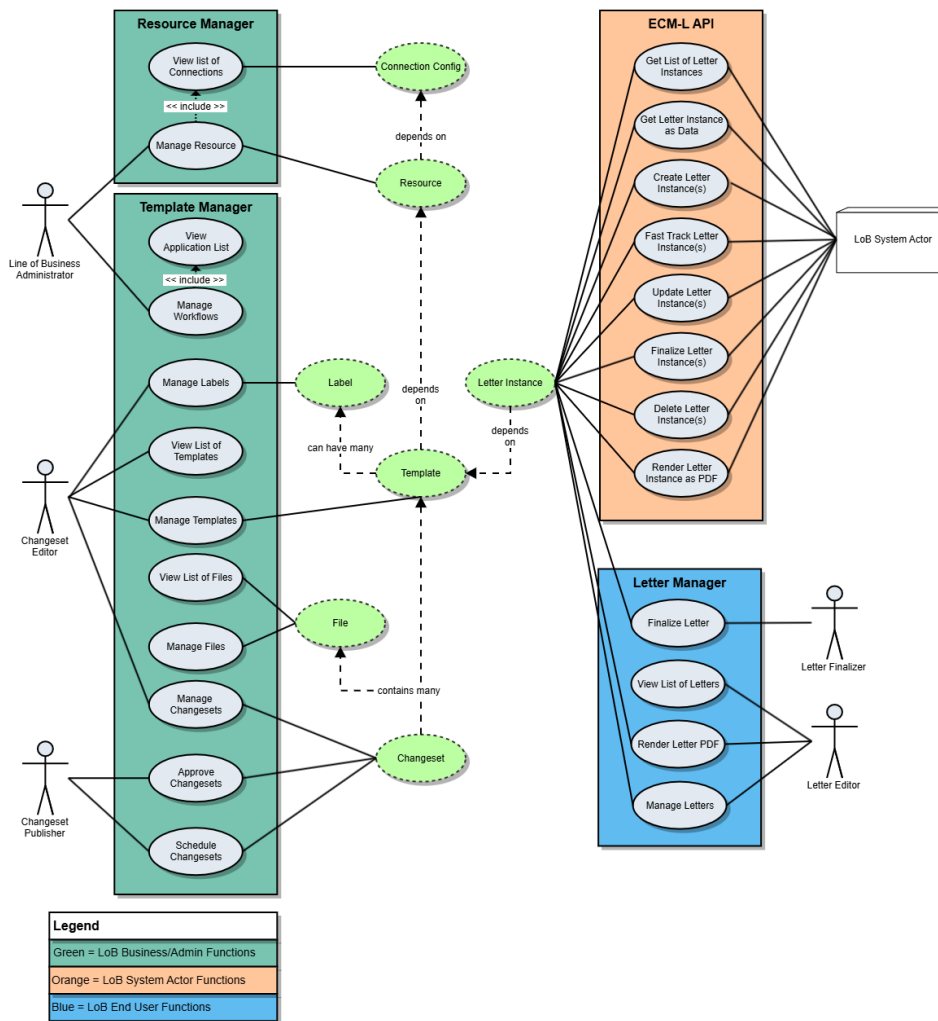
Capability Viewpoint Taxonomy (CV-2)

The following diagram provides an architectural description of the overall, high-level vision and for the capability of this component/tenant.



Use Case

The following diagram visually represents the interactions between users, or actors, and the system. The diagram illustrates the functional requirements of the system.



Main Actors

The following table describes the actors listed in the diagram above. An actor is user in the scope of this architecture and can be a human user or a "system actor" as appropriate.

Name	System	Description

Line of Business Administrator	Resource Manager	A User with the requisite roles and permissions to manage Resources and Connection Configurations.
Line of Business Administrator	Template Manager	A User that manages Workflows, including creating, modifying, and retiring Workflows, configuring them with Key Parameters, and aligning Full Templates to them.
Change set Editor	Template Manager	A User that manages Templates through Changesets, including creating, updating/modifying, testing, recalling, and cancelling Changesets, to facilitate the publication and/or retirement of one or more Template Versions. This user can interact with and manage Template Versions and Retirement Actions in a Changeset, allowing for them to create (specifying a Template's content and configuration), read, update, test, cancel, and retire specific Template Versions. This user can also add Labels to Templates.
Change set Publisher	Template Manager	A User responsible for Approving, Scheduling, and Publishing Changesets.
Line of Business System Actor	ECM-L API	External systems that interact with the ECM-L ecosystem to conduct automated operations such as sending out correspondence without user intervention.
Letter Finalizer	Letter Manager	A user with the ability to Finalize Letters in the Letter Manager.
Letter Editor	Letter Manager	A user with the ability to create, read, update, render, and delete letters in the Letter Manager.

System/Tenant/Application Description

The following table describes each system/tenant/application (rectangles) listed in the diagram above.

System Name	Description
Resource Manager	The tool that allows users to create and Manage Resources and Connection Configurations.
Template Manager	The tool used to manage Changesets, Workflows, Labels and Templates.
ECM-L API	The ECM-L API is the outward face of the ECM-L system and allows for external system actors to interact with the ECM-L system in lieu of a human actor directly interacting with one of the individual components.
Letter Manager	The tool used to manage Letter Instances.

Main User Functions (Use Cases)

The following table describes each of the use cases (ovals) in the diagram above.

Actor (s)	System	Ability	Details
Line of Business Administrator	Resource Manager	Manage Resource	The Line of Business Administrator is able to create, modify, retire, and delete Resources (Data/File), for an Application. The Line of Business Administrator will need to be able to view the list of AVAILABLE Connection Configurations to create Resources. The Line of Business Administrator will need to align or match a given Resource with the Application.
	Template Manager	Manage Workflows	The LOB Admin accesses the Template Manager system and is able to create, modify, and retire Workflows, and add Workflow Key Parameters as well as align Full Templates to those Workflows.
Changeset Editor	Template Manager	Manage Templates	The Changeset Editor accesses the Template Manager to create, modify, and retire Templates. This includes the ability to add or remove Resources for a Template. The Changeset Editor also has the ability to add Inputs to a Template and format the values, as well as add conditional Restrictions to a Template.
		View List of Templates	The Changeset Editor is able to get a full list of all Templates and their respective Versions, for the Application they are working in.

		Manage Files	The Changeset Editor is able to manage, via a Changeset, different types of Files, such as PDFs that can be used as an Attachment and Images, that could be inserted into the content of a Template.
		View List of Files	The Changeset Editor is able to get a full list of all Files and their respective Versions, for the Application they are working in.
		Manage Labels	The Changeset Editor accesses the Template Manager system and is able to create, modify, retire, and delete Template Labels, and add Labels to Templates.
		Manage Changesets	A Changeset Editor is able to create, test, and cancel Changesets.
Changeset Publisher	Template Manager	Approve Changesets	Once a Changeset is in "READY FOR APPROVAL" state, a user with the Changeset Publisher role can Approve the Changeset.
		Schedule Changesets	The Changeset Publisher is able to Schedule Publication of the Changeset immediately or in the future.
Line of Business System Actor	ECM-L API	Get List of Letter Instances	A Line of Business System Actor is able to get Letter Instances based on Application and Key Parameters.
		Get Letter Instance as Data	A Line of Business System Actor is able to get the Letter Instance as a JSON object.
		Create Letter Instance (s)	A Line of Business System Actor is able to create a Letter Instance.
		Fast Track Letter Instance (s)	A Line of Business System Actor creates a Letter Instance and "fast tracks" it through the Letter Instance lifecycle, through Finalization. This makes use of the same operation as the "Create Letter Instance(s)" use case but passing in an optional "autoFinalize" flag.
		Update Letter Instance (s)	A Line of Business System Actor has the ability to make modifications to a Letter Instance(s).
		Finalize Letter Instance (s)	A Line of Business System Actor has the ability to Finalize a Letter Instance(s).
		Delete Letter Instance (s)	A Line of Business System Actor has the ability to soft delete a Letter Instance(s).
		Render Letter Instance (s) as PDF	A Line of Business System Actor is able to Render a Letter as a PDF.
Letter Finalizer	Letter Manager	Finalize Letter	Letter Finalizer has the ability to Finalize a Letter Instance. Finalizing the Letter freezes its content and makes it immutable / unchanging.
Letter Editor	Letter Manager	Manage Letters	The Letter Editor has the ability to create, read, update, and delete a Letter Instance.
		Render Letter as PDF	The Letter Editor is able to get a PDF rendering of a Letter Instance.
		View List of Letters	The ability to view a List of available Letters.

Primary Systems High Level Operational Viewpoint (OV-1)

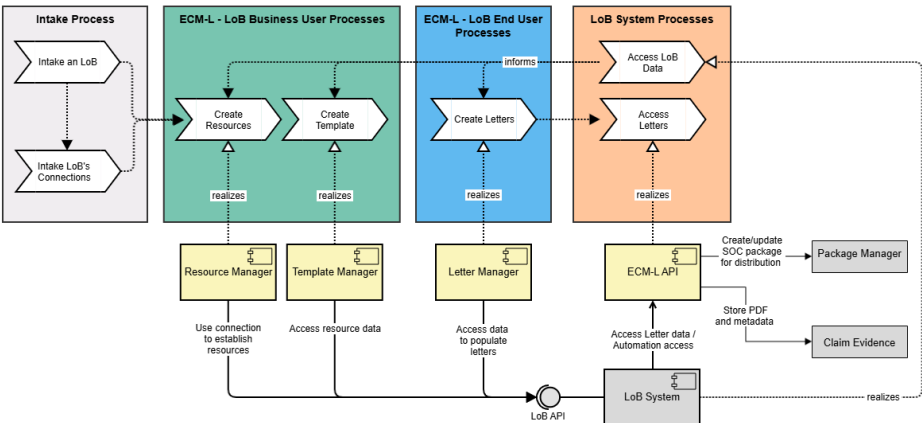
The diagram below is a high level overview of the main architectural components of ECM-L and the functions they facilitate in the value stream.

The first section of the value stream, the intake process, starts as a Jira ticket. This section consists of mainly manual processes provisioning space and storage, establishing user access requirements, and aligning Applications. At this point in the process Line of Business (LoB) data needs will be identified and API connections will be established to accommodate those needs.

The ECM-L LoB Business User Processes box consists of two main values; creating Resources and creating Templates. Resources are assets that will point to LoB Data using the established API connections to populate Letter Instances as needed. The ability to create, test, and manage Resources is realized by the Resource Manager. Creating and managing Templates, or Letter Templates, is realized through the Template Manager.

The ECM-L LoB End User Processes box represents end user interactions with the system to instantiate Letter Instances generated from Templates and populated by user Inputs and Resources.

LoB System Processes encapsulates the ability to access, store, and send Letters as the LoB sees fit. Access to the Letters is capable through the ECM-L API.



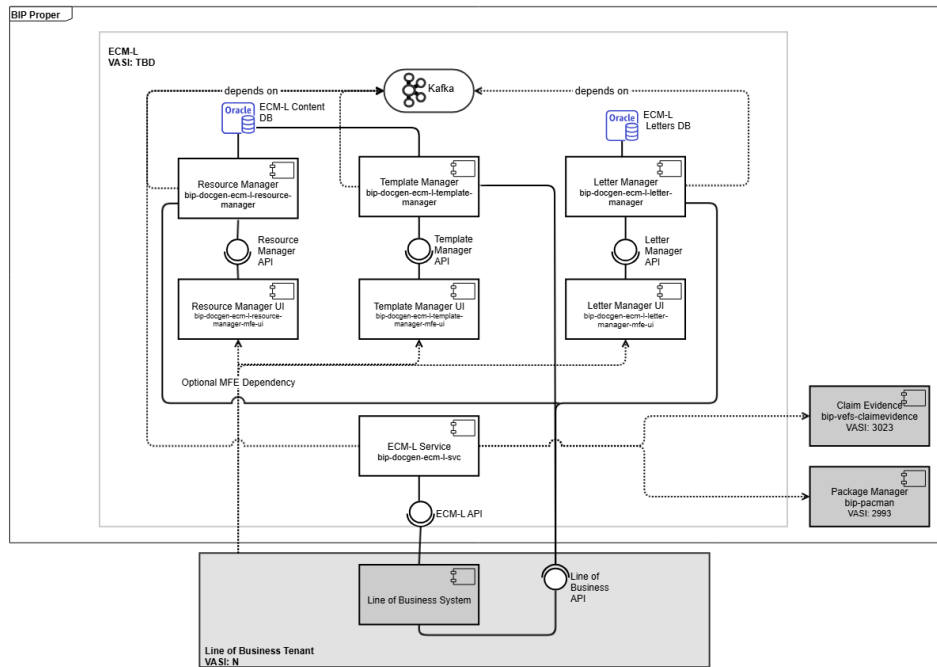
System (VASI)	System Item	Interaction
ECM-L (3000)	Resource Manager	Uses the LoB connection to create and manage Resources.
ECM-L (3000)	Template Manager	Creates and Manages the LoB Templates.
ECM-L (3000)	Letter Manager	Creates and Manages the LoB Letter Instances.
ECM-L (3000)	ECM-L API	Accesses the LoB Data and Letters for the LoB. Letter Instances are able to be stored in Claim Evidence and distributed via Package Manager.
VEFS (3023)	Claim Evidence	Stores the Finalized PDF related to a claim.
PacMan (2993)	Package Manager	Creates a package for mail distribution.
LoB System	LoB System	Uses its connection to establish Resources, Templates and populate Letter Instances with the appropriate data and access the Finalized Letter Instances.

Logical View

The Logical View describes the conceptual view of the functionality and of the systems that will participate. This view defines what nouns are involved and the relationship between them. For example, it may decompose the design into sub-systems and packages, and for significant packages, the decomposition into classes and class utilities. This section may include the Business Object Model, states and transitions for key objects, and other key details about classes and attributes.

Systems Interface Description (SV-1)

The following diagram provides a detailed view of system interfaces, depicting how systems and components interact with each other within the architecture.



Legend

Gray Rectangle = External to core system(s)

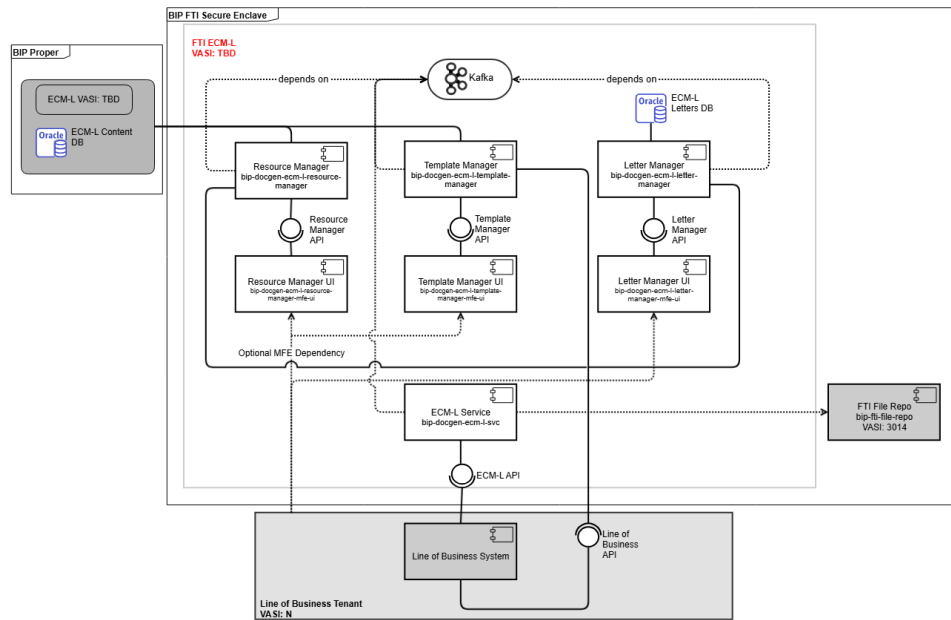
White = Internal to core system(s)

Component References Table

Name	Description	Port	Protocol
ECM-L Content Database	Database containing all ECM-L related data. Database containing "dry" non-contextual data, uncontaminated with PII /PHI/etc.	1521	TCP
ECM-L Letters Database	Database for storage of contextual data containing PII/PHI/etc.	1521	TCP
Resource Manager API	Internal API for the Resource Manager	443	HTTPS /REST
Template Manager API	Internal API for the Template Manager	443	HTTPS /REST
Letter Manager API	Internal API for the Letter Manager	443	HTTPS /REST
ECM-L API	The API for the umbrella service for the ECM-L system, integrated with all of the internal components.	443	HTTPS /REST
Claim Evidence	External API for storage of documents in a Veteran's EFolder.	443	HTTPS /REST
Package Manager	External API for distributing packages of documents from a Veteran's EFolder.	443	HTTPS /REST

FTI Environment Example SV-1

This diagram provides a detailed view of system interfaces with service providers and service consumers, depicting how ECM-L components interact with each other in BIP FTI Secure Enclave for creation and management of FTI-related correspondence. The FTI Secure Enclave is a protected environment within Veterans Benefits platform specifically designed to handle Federal Tax information in compliance with IRS security requirements.



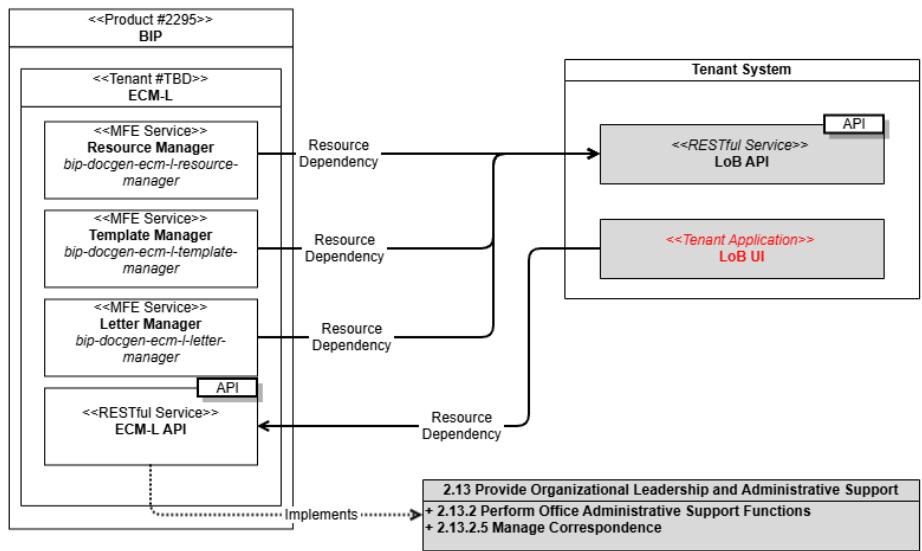
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Component References Table

Name	Description	Port	Protocol
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ECM-L Letters Database	Database containing contextual data contaminated with PII/PHI/etc.	1521	TCP
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Template Manager API	Internal API for the Template Manager	443	HTTPS /REST
Letter Manager API	Internal API for the Letter Manager	443	HTTPS /REST
ECM-L API	The API for the umbrella service for the ECM-L system, integrated with all of the internal components.	443	HTTPS /REST
FTI File Repo	FTI File Repository is the system of record for documents that contain Federal Tax Information and provides API access to those documents.	443	HTTPS /REST

Services Context (SvcV-1) (VASI APIs)

The following diagram describes service compositions and interactions, showing a given Application Components service and immediate service dependencies. The diagram highlights external system services that connect to the focal architecture.

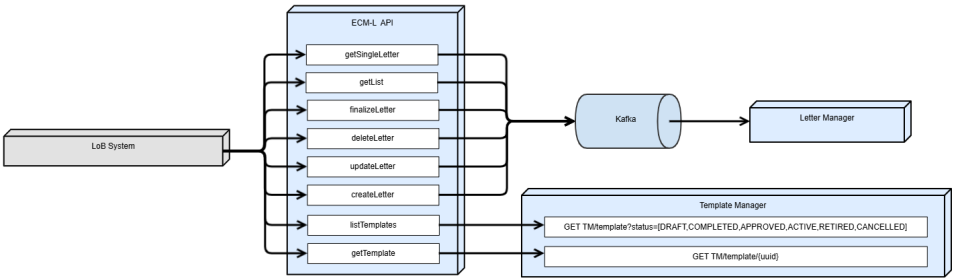


Legend
Gray Rectangle = External to core system(s)
White = Internal to core system(s)

Service / API	Description
Resource Manager	MFE providing LoB with the ability to create, test, replace, and retire Resources.
Template Manager	MFE providing LoB with the ability to create, test and publish Changesets as well as create, test and retire Templates.
Letter Manager	MFE providing ability to create Letter Instances, the Letter Manager will grab the Template and populate using Resources.
ECM-L API	ECM-L API provides the LoB system with access to the Letter Instances for sending and storing. ECM-L API is an outward-facing API for external applications to use to interact with the ECM-L components; it can Render Letter Instances as PDFs as well as handle the storage (in Claim Evidence) and distribution (via Package Manager) of generated PDFs.
LoB API	LoB API is a placeholder for the connection that will be set up for Resources to point to and access data from when populating a Template.
LoB UI	LoB UI will integrate connection to ECM-L MFE's for seamless workflow for the end user.

Services Resource Flow (SvcV-2)

The following diagram specifies the Resource Flows between Services and may also list the protocol stacks used in connections, providing a precise specification of a connection between Services.



Legend
Gray Rectangle = External to core system(s)
White = Internal to core system(s)

HTTP Method	HTTP URL	Operation ID	Performing System	Description
GET	LM/letterInstance/{uuid}	getSingleLetter	Letter Manager	ECM-L API request single Letter Instance
GET	LM/letterInstance	getList	Letter Manager	ECM-L API request of Letter Instances based on the Application and Filters (e.g., workflow, key parameters).
PUT	LM/letterInstance/finalization	finalizeLetter	Letter Manager	ECM-L API ascribes FINAL to an existing Letter Instance that is no longer editable
DELETE	Delete LM/letterInstance	deleteLetter	Letter Manager	ECM-L API deletes a Letter Instance
POST	LM/letterInstance? (autoFinalize=True)	createLetter	Letter Manager	ECM-L API creates a Letter Instance, if autoFinalize is true when the Letter is created it will automatically be Finalized. The autoFinalize enables the ability for LoB system actors to automatically create and then Finalize a Letter
PUT	LM/letterInstance/{uuid}	updateLetter	Letter Manager	ECM-L API updates a Letter Instance
GET	TM/template?status=[DRAFT, COMPLETED, APPROVED, ACTIVE, RETIRED, Canceled]	listTemplates	Template Manager	ECM-L API requests list of Templates
GET	TM/template/{uuid}	getTemplate	Template Manager	ECM-L API requests Template

API Operations Lists

Below are lists of the available operations for the different API's in the ECM-L components.

Resource Manager API Operations

HTTP Method	HTTP URL	Operation ID	Description
GET	/lobs/{lobId}/connection-configurations	getConnectionConfigurationList	Retrieve a list of Connection Configurations for a specific Line of Business (LoB).
GET	/lobs/{lobId}/connection-configurations/{id}	getConnectionConfiguration	Retrieve a specific Connection Configuration by ID for a specific LoB.
PUT	/lobs/{lobId}/connection-configurations/{id}/test	testConnectionConfiguration	Test a specific Connection Configuration by ID for a specific LoB.
PUT	/lobs/{lobId}/connection-configurations/{id}/publish	publishConnectionConfiguration	Publish a specific Connection Configuration by ID for a specific LoB.
PUT	/lobs/{lobId}/connection-configurations/{id}/decommission	decommissionConnectionConfiguration	Decommission a specific Connection Configuration by ID for a specific LoB.
GET	/lobs/{lobId}/resources	getResourceList	Retrieve a list of Resources for a specific Line of Business (LoB).
GET	/lobs/{lobId}/resources/{resourceId}	getResource	Retrieve a Resource by ID for a specific LoB.
GET	/lobs/{lobId}/resources/{resourceId}/versions	getResourceVersionList	Retrieve a list of Resource Versions for a specific LoB.
GET	/lobs/{lobId}/resources/{resourceId}/versions/{id}	getResourceVersion	Retrieve a specific Resource Version for a specific LoB.
PUT	/lobs/{lobId}/resources/{resourceId}/versions/{id}	updateResourceVersion	Update specific Resource Version by ID for a specific LoB.
PUT	/lobs/{lobId}/resources/{resourceId}/versions/{id}/complete	completeResourceVersion	Complete specific Resource Version by ID for a specific LoB.
PUT	/lobs/{lobId}/resources/{resourceId}/versions/{id}/retire	retireResourceVersion	Retire specific Resource Version by ID for a specific LoB.
DELETE	/lobs/{lobId}/resources/{resourceId}/versions/{id}/delete	deleteResourceVersion	Delete specific Resource Version by ID for a specific LoB.
GET	/lobs/{lobId}/resources/{resourceId}/versions/{id}/test	testResourceVersion	Test a Resource Version for a specific LoB.
POST	/lobs/{lobId}/resources/{resourceId}/versions	createResourceVersion	Create a new Resource Version for a specific LoB.
GET	/lobs/{lobId}/key-parameters	getKeyParameterList	Retrieve a list of Key Parameters for a specific LoB.
GET	/lobs/{lobId}/key-parameters/{id}	getKeyParameter	Retrieve a specific Key Parameter by ID for a specific LoB.
PUT	/lobs/{lobId}/key-parameters/{id}	updateKeyParameter	Update specific Key Parameter by ID for a specific LoB.

PUT	/lobs/{lobId}/key-parameters/{id}/retire	retireKeyParameter	Retire specific Key Parameter by ID for a specific LoB.
POST	/lobs/{lobId}/key-parameters	createKeyParameter	Create a new Key Parameter for a specific LoB.

Template Manager API Operations

HTTP Method	HTTP URL	Operation ID	Description
GET	/lobs/{lobId}/applications	getApplicationList	Retrieve a list of Applications for a specific Line of Business (LoB).
GET	/lobs/{lobId}/application/{appld}	getApplication	Retrieve a specific Application by ID for a specific LoB.
GET	/lobs/{lobId}/application/{appld}/workflows	getWorkflowList	Retrieve a list of Workflows for a specific Application.
POST	/lobs/{lobId}/application/{appld}/workflows	createWorkflow	Create a new Workflow for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/workflows/{id}	updateWorkflow	Update a specific Workflow for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/workflows/{id}/retire	retireWorkflow	Retire a specific Workflow for a specific Application.
GET	/lobs/{lobId}/application/{appld}/workflows/{id}	getWorkflow	Retrieve a specific Workflow for a specific Application.
GET	/lobs/{lobId}/application/{appld}/workflows/{id}/template-versions	getWorkflowTemplateVersionList	Retrieve a list of the latest ACTIVE Template Versions for a specific Workflow.
GET	/lobs/{lobId}/application/{appld}/labels	getLabelList	Retrieve a list of Labels for a specific Application.
POST	/lobs/{lobId}/application/{appld}/labels	createLabel	Create a new Label for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/labels/{id}	updateLabel	Update a specific Label for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/labels/{id}/retire	retireLabel	Retire a specific Label for a specific Application.
GET	/lobs/{lobId}/application/{appld}/labels/{id}	getLabel	Retrieve a specific Label for a specific Application. This includes a list of all the Templates that have been assigned this Label.
GET	/lobs/{lobId}/application/{appld}/templates	getTemplateList	Retrieve a list of Templates for a specific Application.
GET	/lobs/{lobId}/application/{appld}/templates/{id}	getTemplate	Retrieve a specific Template for a specific Application.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/history	getTemplateHistory	Retrieve the history for a specific Template for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}	updateTemplate	Update a specific Template for a specific Application.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions	getTemplateVersionList	Retrieve a list of Template Versions for a specific Application.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}	getTemplateVersion	Retrieve a specific Template Version for a specific Application.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/history	getTemplateVersionHistory	Retrieve the history for a specific Template Version for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}	updateTemplateVersion	Update a specific Template Version for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/complete	completeTemplateVersion	Transitions a specific Template Version from DRAFT to COMPLETED.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/approve	approveTemplateVersion	Transitions a specific Template Version from TESTED to APPROVED.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/cancel	cancelTemplateVersion	Transitions a specific Template Version to CANCELED.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/recall	recallTemplateVersion	Recalls a Template Version to DRAFT.
POST	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases	createTestCase	Create a Test Case for a specific Template Version.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases	getTestCaseList	Retrieve a List of Test Cases for a specific Template Version.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}	getTestCase	Retrieve a specific Test Case for a specific Template Version.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/complete	completeTestCase	Move a Test Case into the READY FOR REVIEW state.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/render	renderTestCase	Renders a specific Test Case for a specific Template Version as a PDF.
GET	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{testCaseId}/compare/{id}	compareTestCase	Renders a comparison PDF comparing the rendered PDFs of two specific Test Cases.

PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/pass	passTestCase	Move a Test Case into the PASSED state.
DELETE	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/delete	deleteTestCase	Move a Test Case into the DELETED state.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/fail	failTestCase	Move a Test Case into the FAILED state.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/test-cases/{id}/recall	recallTestCase	Recall a Test Case into the DRAFT state.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/complete	completeTemplateVersion	Transitions a specific Template Version from DRAFT to COMPLETED.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/approve	approveTemplateVersion	Transitions a specific Template Version from TESTED to APPROVED.
PUT	/lobs/{lobId}/application/{appld}/templates/{id}/versions/{id}/recall	recallTemplateVersion	Recalls a specific Template Version into the DRAFT state.
GET	/lobs/{lobId}/application/{appld}/changesets	getChangesetList	Retrieve a list of Changesets for a specific Application.
POST	/lobs/{lobId}/application/{appld}/changesets	createChangeset	Create a Changeset for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}	updateChangeset	Update a specific Changeset for a specific Application.
GET	/lobs/{lobId}/application/{appld}/changesets/{id}	getChangeset	Retrieve a specific Changeset for a specific Application.
GET	/lobs/{lobId}/application/{appld}/changesets/{id}/history	getChangesetHistory	Retrieve the history for a specific Changeset for a specific Application.
GET	/lobs/{lobId}/application/{appld}/changesets/{id}/template-versions	getChangesetTemplateVersionList	Get the Template Versions for a specific Changeset.
POST	/lobs/{lobId}/application/{appld}/changesets/{id}/template-versions	createTemplateVersion	Create a Template Version for a specific Changeset.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/cancel	cancelChangeset	Cancels a specific Changeset for a specific Application.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/complete	completeChangeset	Transitions a specific Changeset for a specific Application into the READY TO TEST state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/submit	submitChangeset	Transitions a specific Changeset for a specific Application into the READY FOR APPROVAL state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/reject	rejectChangeset	Transitions a specific Changeset for a specific Application into the REJECTED state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/approve	approveChangeset	Transitions a specific Changeset for a specific Application into the APPROVED state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/schedule	scheduleChangeset	Transitions a specific Changeset for a specific Application into the SCHEDULED state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/recall	recallChangeset	Recalls a specific Changeset for a specific Application into the DRAFT state.
PUT	/lobs/{lobId}/application/{appld}/changesets/{id}/publish	publishChangeset	Transitions a specific Changeset for a specific Application into the PUBLISHED state.

Letter Manager API Operations

HTTP Method	HTTP URL	Operation ID	Description
POST	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances	createLetterInstance	Instantiates a Letter Instance based on a Template Version given the Key Parameters and the Template Version within the request body. Can optionally 'fast-track' the created Letter Instance all the way through the Finalization process.
POST	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/search	searchLetterInstances	Returns a search response including a list of Letter Instances related to a set of given Key Parameters within the request body.
GET	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}	getLetterInstance	Retrieves a specific Letter Instance within a specific Workflow.
GET	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/history	getLetterInstanceHistory	Retrieves the history for a specific Letter Instance within a specific Workflow.
GET	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/render	renderLetterInstance	Renders a specific Letter Instance within a specific Workflow.

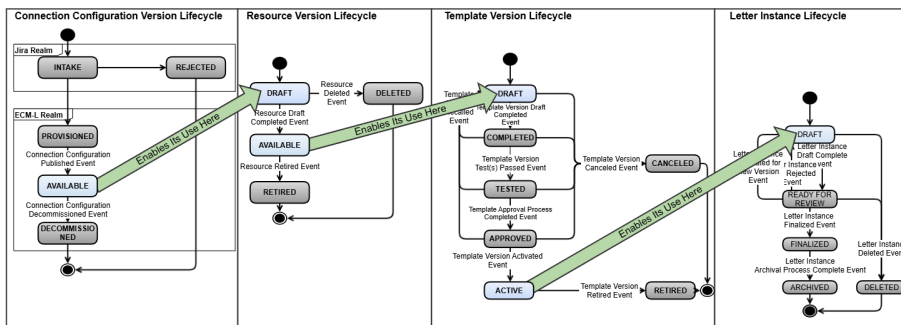
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}	updateLetterInstance	Updates a specific Letter Instance within a specific Workflow.
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/restantiate	restantiateLetterInstance	Reinstantiates a Letter Instance using the current / most recent ACTIVE Template Version for it's Template.
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/submit	submitLetterInstance	Submits a specific DRAFT Letter Instance within a specific Workflow for review, updating it into the READY FOR REVIEW state.
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/recall	recallLetterInstance	Recalls a specific Letter Instance back to DRAFT after it was submitted for review.
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/reject	rejectLetterInstance	Rejects a specific DRAFT Letter Instance within a specific Workflow after review.
PUT	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/finalize	finalizeLetterInstance	Finalizes a Letter Instance, updating it into the FINALIZED state. For Applications configured to Store and Distribute Letters, this will also result in the Storage and Distribution of a Rendered PDF based on the Finalized Letter Instance.
DELETE	/lobs/{lobId}/application/{appld}/workflow/{workflowId}/letter-instances/{id}/delete	deleteLetterInstance	Updates the status of a Letter Instance to DELETED.

State Transitions (OV-6b)

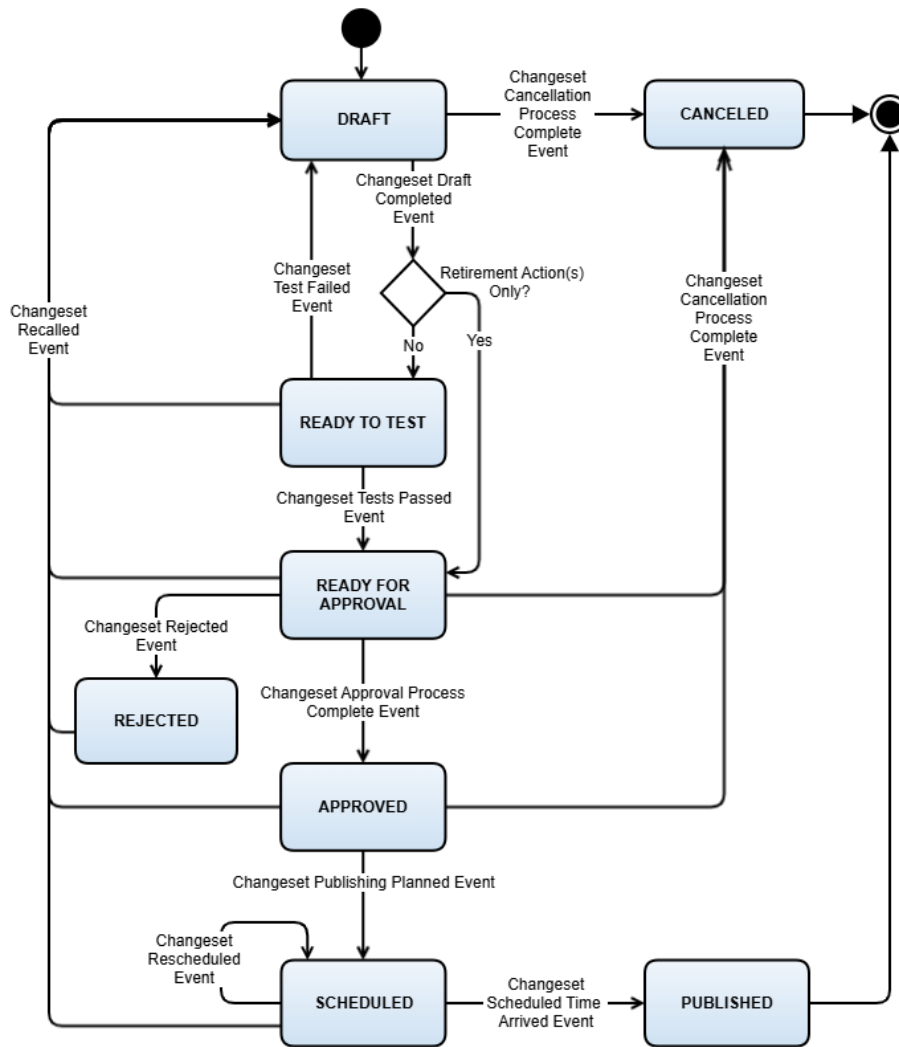
The following diagrams map out the various states of being for individual ECM-L elements and the flow patterns between those states, as well any events that are triggered as these elements transition from one state to the next. These state transition diagrams represent the "heartbeat" of the ECM-L ecosystem and are grounded in an event-based architectural philosophy.

Lifecycle Interactions

This diagram helps to outline the states in which an element of the ECM-L ecosystem becomes available for the elements that are able to use them.



Changeset Lifecycle



DRAFT

This Changeset has been created, named, and work has begun to prepare it for publication. During this state, Templates are being created, added/removed to/from the Changeset, and tested. During this state the Changeset and its content remain modifiable. Test Cases can be created and executed to allow for incremental testing of a Template, but can not yet be marked as PASSED.

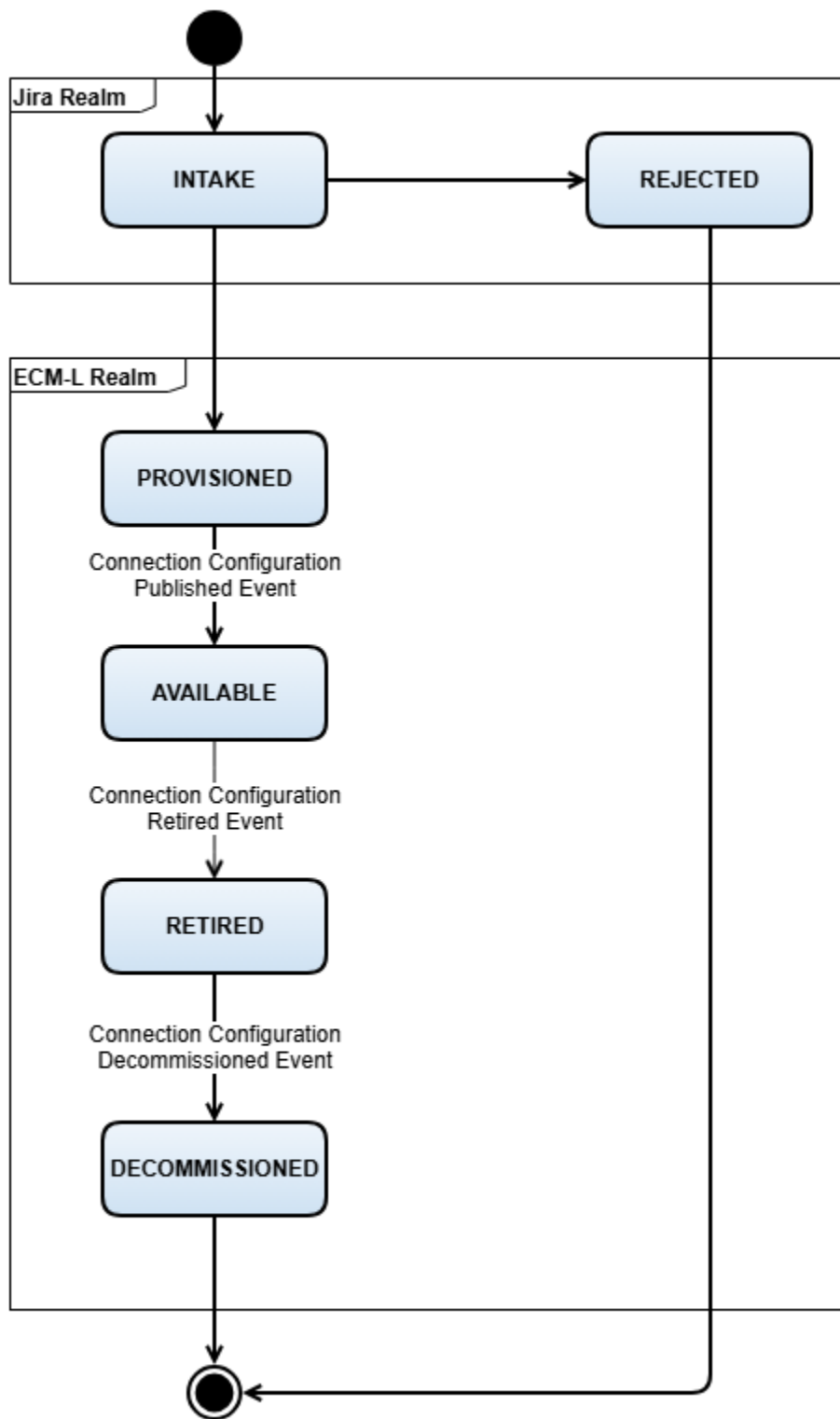
Transitions from this state:

EVENT	TO	DESCRIPTION
ChangesetDraftCompletedEvent	READY TO TEST	All Template Versions have been marked as Complete and the Changeset Editor has indicated that the Changeset is ready to be tested.
ChangesetCancellationProcessCompleteEvent	CANCELED	The Changeset and all of its contents have been Canceled by the Changeset Editor and will no longer be moved towards publication.

READY TO TEST	This Changeset and all of the work within its contents has been completed, there are Template Versions within the Changeset that need to be marked as Passed, and the Changeset Editor has indicated that the Changeset is READY TO TEST. From this state, Template Versions must each have at least one Passed Test Case before the Changeset they contain can be submitted for approval. This Status is skipped for a Changeset that contains only Retirement Actions, since Retirement Actions are not tested. The Changeset Editor can continue to create and modify Test Cases, run them, and mark them as PASSED. The content of the Template Versions can not be modified unless the Changeset is Recalled back to the DRAFT state. Transitions from this state:															
	<table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>ChangesetTestsPassedEvent</td><td>READY FOR APPROVAL</td><td>All Test Cases have been created and have PASSED, and Changeset Editor has submitted the Changeset for Approval. Marking a Test Case as PASSED means the Changeset Editor has created a Test Case, executed it, viewed the results, and verified that the Rendered PDF looks as expected. The Changeset Editor can create multiple Test Cases with different data as needed, however all must be marked as PASSED before this transition can take place.</td></tr><tr><td>ChangesetRecalledEvent</td><td>DRAFT</td><td>This Changeset has been recalled to allow for additional modification.</td></tr></table>	EVENT	TO	DESCRIPTION	ChangesetTestsPassedEvent	READY FOR APPROVAL	All Test Cases have been created and have PASSED, and Changeset Editor has submitted the Changeset for Approval. Marking a Test Case as PASSED means the Changeset Editor has created a Test Case, executed it, viewed the results, and verified that the Rendered PDF looks as expected. The Changeset Editor can create multiple Test Cases with different data as needed, however all must be marked as PASSED before this transition can take place.	ChangesetRecalledEvent	DRAFT	This Changeset has been recalled to allow for additional modification.						
	EVENT	TO	DESCRIPTION													
	ChangesetTestsPassedEvent	READY FOR APPROVAL	All Test Cases have been created and have PASSED, and Changeset Editor has submitted the Changeset for Approval. Marking a Test Case as PASSED means the Changeset Editor has created a Test Case, executed it, viewed the results, and verified that the Rendered PDF looks as expected. The Changeset Editor can create multiple Test Cases with different data as needed, however all must be marked as PASSED before this transition can take place.													
ChangesetRecalledEvent	DRAFT	This Changeset has been recalled to allow for additional modification.														
READY FOR APPROVAL	Every Template Version in the Changeset has at least one corresponding Test Case that has been marked as Passed, and a Changeset Editor has submitted the Changeset for Approval. For Changesets that do not contain a Template Version, such as Changesets that only contain a Retirement Action, no Test Cases are required. While within this state Test Cases can continue to be created, executed, and marked as PASSED to allow the Changeset Publisher to create and run additional Test Cases. Prior to Approving the Changeset, all of the contents of the Changeset, such as the Template Versions and Retirement Actions, must be individually Approved. Transitions from this state:															
	<table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>ChangesetApprovalProcessCompletedEvent</td><td>APPROVED</td><td>The Changeset and its contents has moved through the applicable Approval processes and has been APPROVED by a Changeset Publisher.</td></tr><tr><td>ChangesetRecalledEvent</td><td>DRAFT</td><td>This Changeset has been recalled by a Changeset Publisher to allow for additional modification.</td></tr><tr><td>ChangesetRejectedEvent</td><td>REJECTED</td><td>Changeset Publisher reviewed and REJECTED the Changeset. Changeset Publisher must provide a Rejection Reason before the Changeset can transition to REJECTED state.</td></tr><tr><td>ChangesetCancellationProcessCompleteEvent</td><td>CANCELED</td><td>The Changeset and all of its contents have been CANCELED by a Changeset Publisher and will no longer be moved towards publication.</td></tr></table>	EVENT	TO	DESCRIPTION	ChangesetApprovalProcessCompletedEvent	APPROVED	The Changeset and its contents has moved through the applicable Approval processes and has been APPROVED by a Changeset Publisher.	ChangesetRecalledEvent	DRAFT	This Changeset has been recalled by a Changeset Publisher to allow for additional modification.	ChangesetRejectedEvent	REJECTED	Changeset Publisher reviewed and REJECTED the Changeset. Changeset Publisher must provide a Rejection Reason before the Changeset can transition to REJECTED state.	ChangesetCancellationProcessCompleteEvent	CANCELED	The Changeset and all of its contents have been CANCELED by a Changeset Publisher and will no longer be moved towards publication.
	EVENT	TO	DESCRIPTION													
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	ChangesetRecalledEvent	DRAFT	This Changeset has been recalled by a Changeset Publisher to allow for additional modification.													
	ChangesetRejectedEvent	REJECTED	Changeset Publisher reviewed and REJECTED the Changeset. Changeset Publisher must provide a Rejection Reason before the Changeset can transition to REJECTED state.													
ChangesetCancellationProcessCompleteEvent	CANCELED	The Changeset and all of its contents have been CANCELED by a Changeset Publisher and will no longer be moved towards publication.														
REJECTED	This Changeset Publisher has reviewed the contents of the Changeset and REJECTED it. They provide a Rejection Reason so that the Changeset Editor that owns the Changeset can review it and return it to the DRAFT state for additional modification. Transitions from this state:															
	<table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>ChangesetRecalledEvent</td><td>DRAFT</td><td>The Changeset Editor, after reviewing the REJECTED Changeset, moves it back to the DRAFT state for additional modification..</td></tr></table>	EVENT	TO	DESCRIPTION	ChangesetRecalledEvent	DRAFT	The Changeset Editor, after reviewing the REJECTED Changeset, moves it back to the DRAFT state for additional modification..									
EVENT	TO	DESCRIPTION														
ChangesetRecalledEvent	DRAFT	The Changeset Editor, after reviewing the REJECTED Changeset, moves it back to the DRAFT state for additional modification..														

APPROVED	This Changeset Publisher has reviewed the contents of the Changeset and Approved it. All of the contents of the Changeset must be Approved before the Changeset itself can be APPROVED. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>ChangesetRecalledEvent</td><td>DRAFT</td><td>Changeset Publisher has determined that the APPROVED Changeset needs more work and so it is returned to the DRAFT state for additional modification.</td></tr><tr><td>ChangesetCancellationProcessCompleteEvent</td><td>CANCELED</td><td>The Changeset and all of its contents have been CANCELED by Changeset Publisher and will no longer be moved towards publication.</td></tr><tr><td>ChangesetPublishingPlannedEvent</td><td>SCHEDULED</td><td>The Changeset is set up to be Published, either immediately or for a specific date.</td></tr></table>	EVENT	TO	DESCRIPTION	ChangesetRecalledEvent	DRAFT	Changeset Publisher has determined that the APPROVED Changeset needs more work and so it is returned to the DRAFT state for additional modification.	ChangesetCancellationProcessCompleteEvent	CANCELED	The Changeset and all of its contents have been CANCELED by Changeset Publisher and will no longer be moved towards publication.	ChangesetPublishingPlannedEvent	SCHEDULED	The Changeset is set up to be Published, either immediately or for a specific date.
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ChangesetPublishingPlannedEvent	SCHEDULED	The Changeset is set up to be Published, either immediately or for a specific date.											
SCHEDULED	This Changeset Publisher has Scheduled it to be PUBLISHED at the Scheduled Date. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>ChangesetRecalledEvent</td><td>DRAFT</td><td>Changeset Publisher has determined that the SCHEDULED Changeset needs more work and so it is returned to the DRAFT state for additional modification.</td></tr><tr><td>ChangesetRescheduledEvent</td><td>SCHEDULED</td><td>Changeset Publisher has changed the Scheduled Date, which emits a Changeset Rescheduled Event.</td></tr><tr><td>ChangesetScheduledTimeArrivedEvent</td><td>PUBLISHED</td><td>The Scheduled time set for Publication has arrived and the Changeset is PUBLISHED and ready for use. This transition implies the transition of all new and included Templates to the ACTIVE state and any ACTIVE Templates targeted by a Retirement Action will transition to RETIRED.</td></tr></table>	EVENT	TO	DESCRIPTION	ChangesetRecalledEvent	DRAFT	Changeset Publisher has determined that the SCHEDULED Changeset needs more work and so it is returned to the DRAFT state for additional modification.	ChangesetRescheduledEvent	SCHEDULED	Changeset Publisher has changed the Scheduled Date, which emits a Changeset Rescheduled Event.	ChangesetScheduledTimeArrivedEvent	PUBLISHED	The Scheduled time set for Publication has arrived and the Changeset is PUBLISHED and ready for use. This transition implies the transition of all new and included Templates to the ACTIVE state and any ACTIVE Templates targeted by a Retirement Action will transition to RETIRED.
EVENT	TO	DESCRIPTION											
ChangesetRecalledEvent	DRAFT	Changeset Publisher has determined that the SCHEDULED Changeset needs more work and so it is returned to the DRAFT state for additional modification.											
ChangesetRescheduledEvent	SCHEDULED	Changeset Publisher has changed the Scheduled Date, which emits a Changeset Rescheduled Event.											
ChangesetScheduledTimeArrivedEvent	PUBLISHED	The Scheduled time set for Publication has arrived and the Changeset is PUBLISHED and ready for use. This transition implies the transition of all new and included Templates to the ACTIVE state and any ACTIVE Templates targeted by a Retirement Action will transition to RETIRED.											
PUBLISHED	This Changeset is now live and effective. All of the related Templates Versions are moved into the ACTIVE and all of the related Retirement Actions are actualized by moving their respective Template Versions to the RETIRED status. ACTIVE Templates are available to create Letter Instances.												
CANCELED	This Changeset was deemed unnecessary and has been CANCELED. A PUBLISHED Changeset cannot be CANCELED.												

Connection Configuration Version Lifecycle



Jira Realm

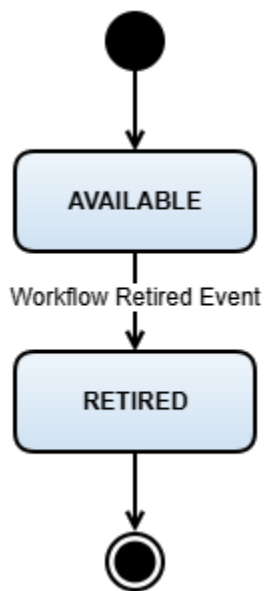
These are not statuses that would surface within the ECM-L eco-system because they represent activity outside of it. Events do not take place here since these activities are not a part of the ECM-L components.

INTAKE	Dev Ops is processing the necessary intake documentation submitted by a Line of Business (LoB) to configure the ECM-L eco-system to interact with and integrate with the API submitted for the Connection Configuration.	
	Transitions from this state:	
	TO	DESCRIPTION
	REJECTED	Something went wrong during the processing of the Intake Ticket and Line of Business (LoB) intervention is now needed to correct the information in the ticket.
REJECTED	PROVISIONED	The Systems or Operations Teams have completed what is necessary to validate the integration with the Connection Configuration's targeted API. The PROVISIONED Connection Configuration will be visible in the Resource Manager.
	The intake process was unsuccessful. The Line of Business and Dev Ops teams should work together to determine what caused the integration failure and work to address it.	

ECM-L Realm

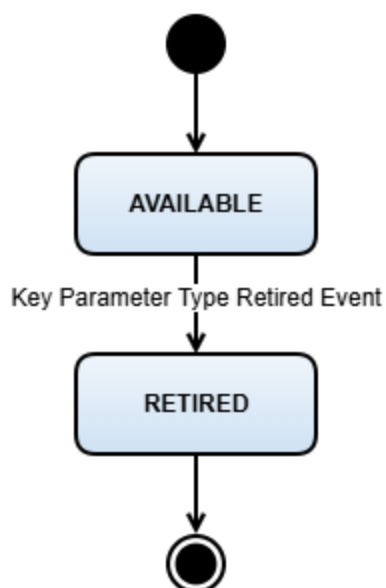
PROVISIONED	The Connection Configuration has made it through the intake process and is ready for a Line of Business Administrator to test it and make it AVAILABLE for use.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	ConnectionConfigurationPublishedEvent	AVAILABLE	The LoB Administrator has tested the PROVISIONED Connection and successfully connected from within the Resource Manager. The LOB Administrator has changed the Connection from PROVISIONED to AVAILABLE.
AVAILABLE	The Connection Configuration has been properly PROVISIONED and Tested, and the LoB Administrator has made it AVAILABLE for use. While in this state the Connection Configuration is AVAILABLE to be used by Resources.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	ConnectionConfigurationRetiredEvent	RETIRED	An LoB Administrator has indicated that the Connection Configuration can no longer be used by the Resources configured to connect to it.
RETIRED	No longer available for use by anything new.		
	The Connection Configuration Version can no longer be used in the configuration of new Resources.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	ConnectionConfigurationDecommissionedEvent	DECOMMISSIONED	An LoB Administrator will mark the Connection Configuration as Decommissioned. As a result of the decommissioning, the Connection will no longer respond/provide data to the Resources configured to connect to it and no new Resources can be configured to connect.
DECOMMISSIONED	The API associated with this Connection Configuration is non-operational and can no longer respond to requests.		

Workflow Lifecycle



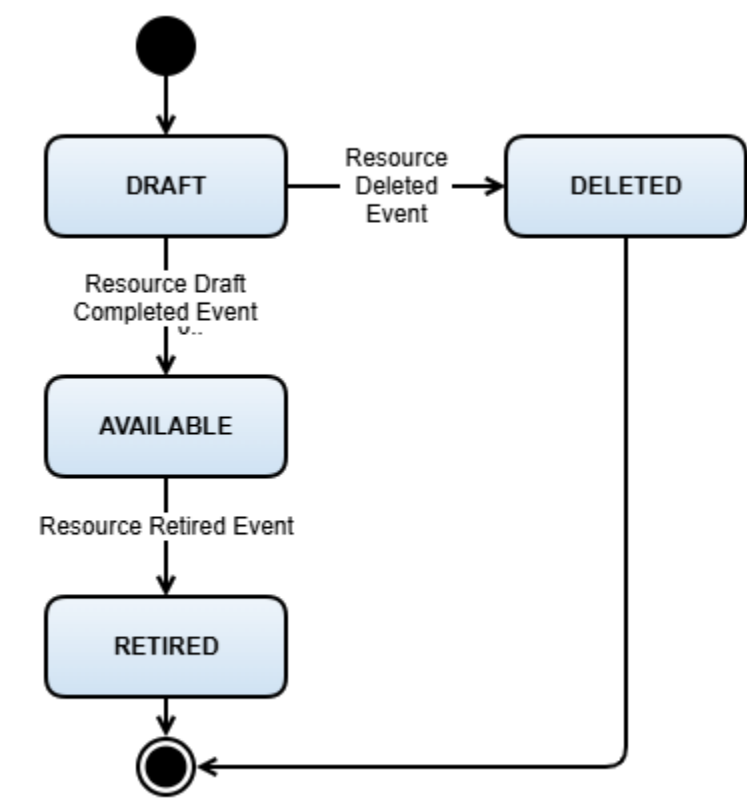
AVAILABLE	This Workflow has been created by a Line of Business (LoB) Administrator and is AVAILABLE for use. In this state Templates can be aligned to the Workflow and Letter Instances can be created within it.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
RETIRED	WorkflowRetiredEvent	RETIRED	Workflow is no longer needed and RETIRED from use.
	No longer available for use by anything new. No new Letter Instances can be created within this Workflow, and no more Templates can be aligned to it.		

Key Parameter Type Lifecycle



AVAILABLE	This Key Parameter Type has been added into the Resource Manager by a Line of Business (LoB) Administrator and is now AVAILABLE for use by Workflows and Resources.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	KeyParameterTypeRetired Event	RETIR ED	Key Parameter Type is no longer needed and has been RETIRED by a LoB Administrator.
RETIRED	No longer available for use by anything new.		
	This Key Parameter Type can no longer be selected as an option when choosing one during the configuration of a Workflow or a Resource.		

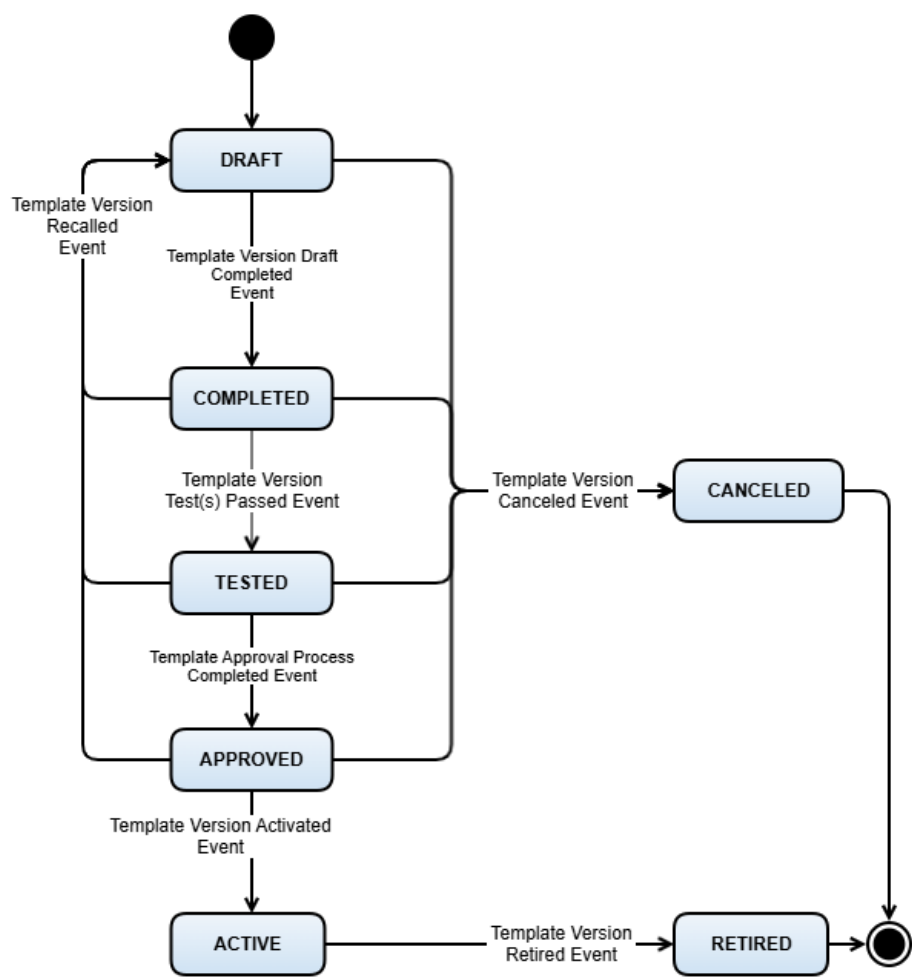
Resource Version Lifecycle



DRAFT	This Resource has been created by a Line of Business (LoB) Administrator using an AVAILABLE Connection Configuration and is in the process of configuring it.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	ResourceDeletedEvent	DELET ED	The Resource was never made AVAILABLE for use, is no longer needed, and has been DELETED by an LoB Administrator.
	ResourceDraftComple tedEvent	AVAILA BLE	The Resource has been completely configured and a LoB Administrator has made it AVAILABLE for use.

AVAILABLE	This Resource is fully configured and a LoB Administrator has made it AVAILABLE for use in Template Versions.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	ResourceRetiredEvent	RETIRED	Resource is no longer needed, but has already been made AVAILABLE for use, so it is RETIRED.
RETIRED	No longer available for use by anything new. This Resource can no longer be used in the configuration of a new Template Version.		
DELETED	This Resource was never made AVAILABLE for use, is no longer needed, and has been DELETED.		

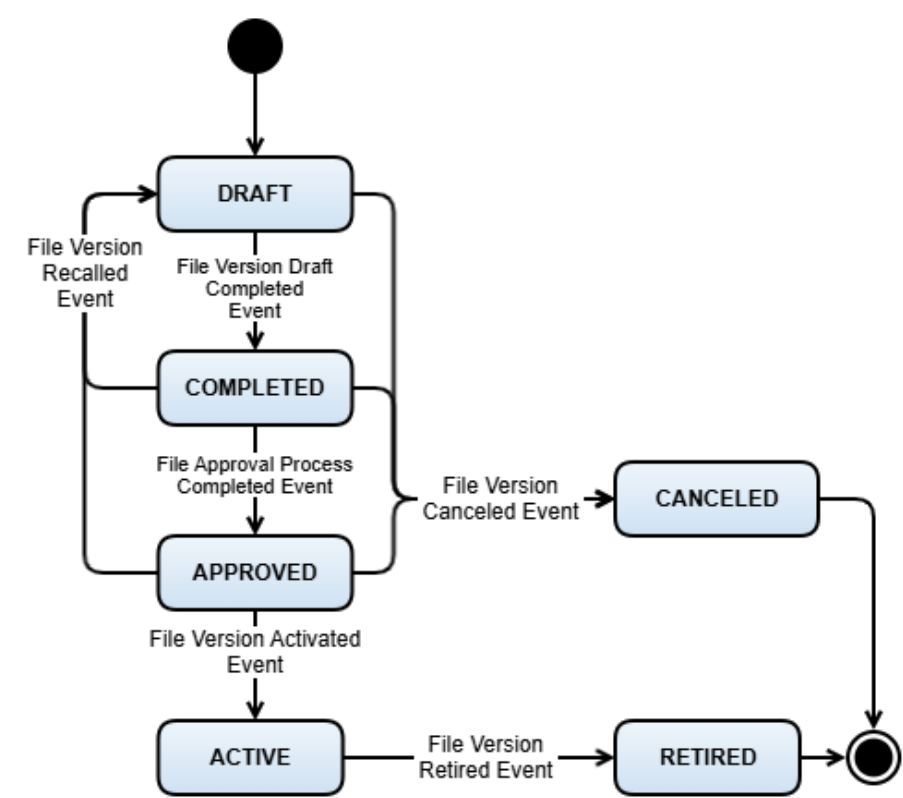
Template Version Lifecycle



DRAFT	This Template Version has been created, named, and is editable. Entering this state, an incremented Version number is assigned. Version numbers are unique to each individual Template Version. This is the only state in which the content and configuration of a Template Version can be modified. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TemplateVersionDraftCompletedEvent</td><td>COMPLETED</td><td>The Changeset Editor is satisfied with their modifications to the Template Version and has marked it as COMPLETED.</td></tr><tr><td>TemplateVersionCanceledEvent</td><td>CANCELED</td><td>This Template Version is no longer needed and has been CANCELED.</td></tr></table>	EVENT	TO	DESCRIPTION	TemplateVersionDraftCompletedEvent	COMPLETED	The Changeset Editor is satisfied with their modifications to the Template Version and has marked it as COMPLETED.	TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.			
EVENT	TO	DESCRIPTION											
TemplateVersionDraftCompletedEvent	COMPLETED	The Changeset Editor is satisfied with their modifications to the Template Version and has marked it as COMPLETED.											
TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.											
COMPLETED	This Template Version's updates and modifications are finished and the Changeset Editor has marked it as COMPLETED. The content and the configuration of the Template Version can no longer be modified. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TemplateVersionRecalledEvent</td><td>DRAFT</td><td>The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.</td></tr><tr><td>TemplateVersionTestPassedEvent</td><td>TESTED</td><td>All of the Test Cases for this Template Version have been run and have been marked as PASSED.</td></tr><tr><td>TemplateVersionCanceledEvent</td><td>CANCELED</td><td>This Template Version is no longer needed and has been CANCELED.</td></tr></table>	EVENT	TO	DESCRIPTION	TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.	TemplateVersionTestPassedEvent	TESTED	All of the Test Cases for this Template Version have been run and have been marked as PASSED.	TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.
EVENT	TO	DESCRIPTION											
TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.											
TemplateVersionTestPassedEvent	TESTED	All of the Test Cases for this Template Version have been run and have been marked as PASSED.											
TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.											
TESTED	This Template Version's Test Cases have all been reviewed and marked as PASSED by a Changeset Editor. New Test Cases can be added to this Template Version while the Changeset is in the Ready for Approval state if the Changeset Publisher decides to add in additional testing during the approval process. In this state the content and configuration of the Template Version can no longer be modified. If a user wishes to make additional modifications to it they must Recall it back to the DRAFT state. <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TemplateVersionApprovalProcessCompletedEvent</td><td>APPROVED</td><td>The Template Version has gone through the applicable approval process and been APPROVED.</td></tr><tr><td>TemplateVersionRecalledEvent</td><td>DRAFT</td><td>The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.</td></tr></table>	EVENT	TO	DESCRIPTION	TemplateVersionApprovalProcessCompletedEvent	APPROVED	The Template Version has gone through the applicable approval process and been APPROVED.	TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.			
EVENT	TO	DESCRIPTION											
TemplateVersionApprovalProcessCompletedEvent	APPROVED	The Template Version has gone through the applicable approval process and been APPROVED.											
TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.											
APPROVED	This Template Version has passed testing and been approved. Test Cases can no longer be added to this Template Version. In this state it can no longer be modified. If a user wishes to make additional modifications to it they must Recall it back to the DRAFT state. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TemplateVersionActivatedEvent</td><td>ACTIVE</td><td>This Template Version was in a Changeset that has been PUBLISHED.</td></tr><tr><td>TemplateVersionCanceledEvent</td><td>CANCELED</td><td>This Template Version is no longer needed and has been CANCELED.</td></tr><tr><td>TemplateVersionRecalledEvent</td><td>DRAFT</td><td>The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.</td></tr></table>	EVENT	TO	DESCRIPTION	TemplateVersionActivatedEvent	ACTIVE	This Template Version was in a Changeset that has been PUBLISHED.	TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.	TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.
EVENT	TO	DESCRIPTION											
TemplateVersionActivatedEvent	ACTIVE	This Template Version was in a Changeset that has been PUBLISHED.											
TemplateVersionCanceledEvent	CANCELED	This Template Version is no longer needed and has been CANCELED.											
TemplateVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this Template Version needs more work and moves it back to DRAFT for further modification.											

ACTIVE	This Template Version has been published and is live and available for the instantiation of Letter Instances.		
	In this state it can no longer be modified. It can also no longer be Recalled.		
	If a user wishes to make additional modifications to it they must create a new Version.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	TemplateVersionRetiredEvent	RETIRED	A Retirement Action targeting this Template Version has been published through a Changeset that results in this Template Version moving into a RETIRED state.
CANCELED	This Template Version is no longer needed and has been CANCELED.		
RETIRED	No longer available for use by anything new.		
	This Template Version can no longer be used to instantiate new Letter Instances.		

File Version Lifecycle



DRAFT

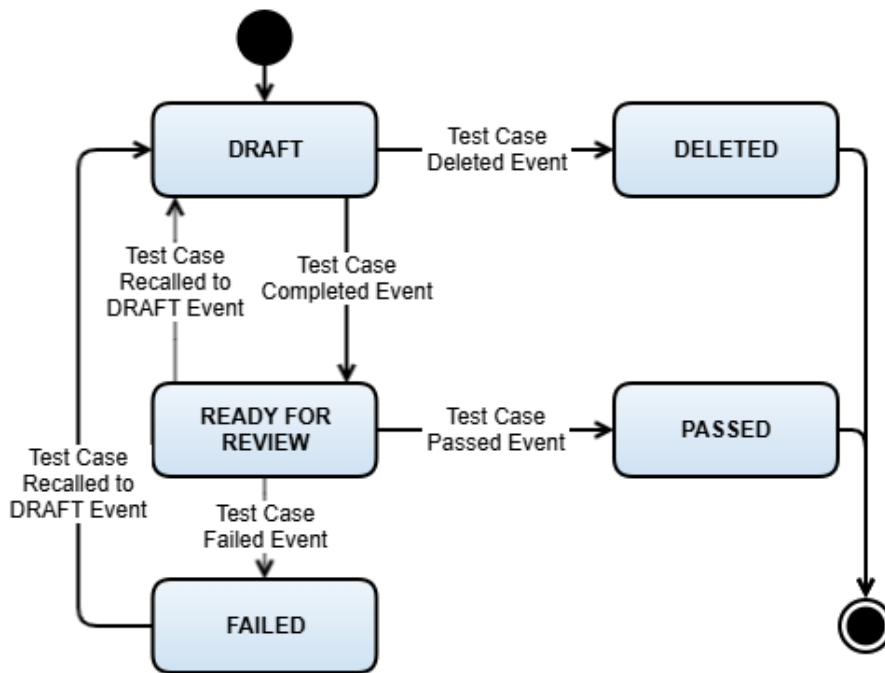
This File Version has been created, named, and is editable. Entering this state, an incremented Version number is assigned. Version numbers are unique to each individual File Version. This is the only state in which the content and configuration of a File Version can be modified.

Transitions from this state:

EVENT	TO	DESCRIPTION
FileVersionDraftCompletedEvent	COMPLETED	The Changeset Editor is satisfied with their modifications to the File Version and has marked it as COMPLETED.
FileVersionCanceledEvent	CANCELED	This File Version is no longer needed and has been CANCELED.

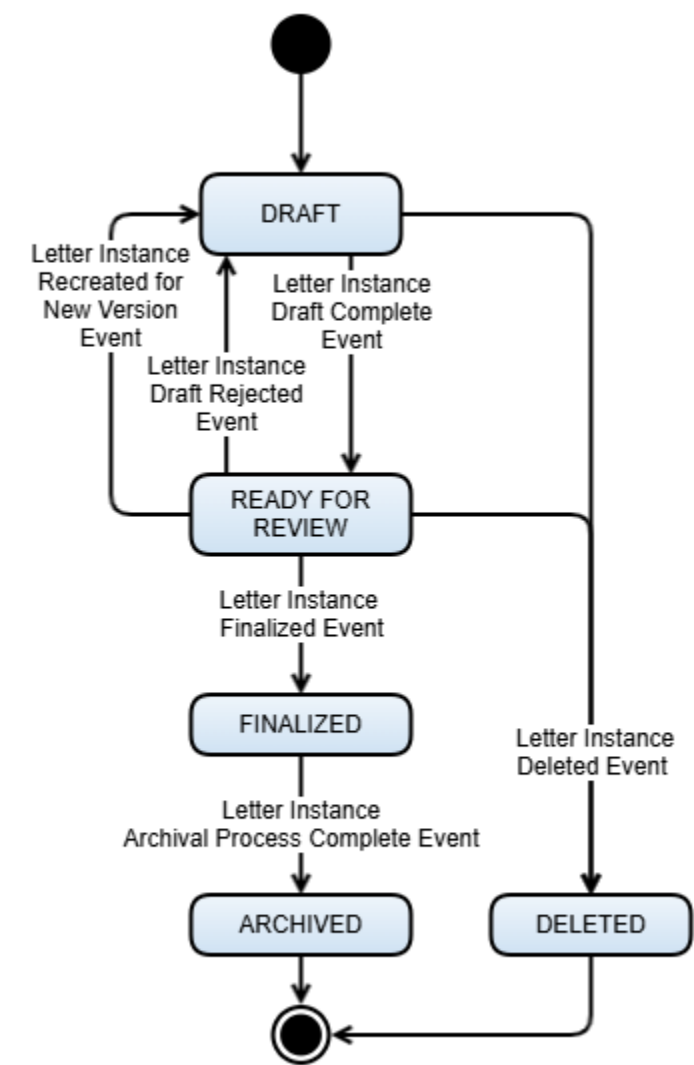
COMPLETED	This File Version's updates and modifications are finished and the Changeset Editor has marked it as COMPLETED.		
	In this state it can no longer be modified. If a user wishes to make additional modifications to it they must Recall it back to the DRAFT state.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	FileVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this File Version needs more work and moves it back to DRAFT for further modification.
	FileVersionCancelledEvent	CANCELLED	This File Version is no longer needed and has been CANCELED.
APPROVED	This File Version has been approved.		
	In this state it can no longer be modified. If a user wishes to make additional modifications to it they must Recall it back to the DRAFT state.		
	Note: Unlike Template Versions, there is no need for testing a File Version.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	FileVersionActivatedEvent	ACTIVE	This File Version was in a Changeset that has been PUBLISHED.
	FileVersionCancelledEvent	CANCELLED	This File Version is no longer needed and has been CANCELED.
	FileVersionRecalledEvent	DRAFT	The Changeset Editor has decided that this File Version needs more work and moves it back to DRAFT for further modification.
ACTIVE	This File Version has been published and is available for use.		
	In this state it can no longer be modified. It can also no longer be Recalled.		
	If a user wishes to make additional modifications to it they must create a new Version.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	FileVersionRetiredEvent	RETIRED	A Retirement Action targeting this File Version has been published through a Changeset that results in this File Version moving into a RETIRED state.
CANCELED	This File Version is no longer needed and has been CANCELED.		
RETIRED	No longer available for use by anything new.		
	This File Version can no longer be used in new Template Versions.		

Test Case Lifecycle



DRAFT	A Changeset Editor or Changeset Publisher has created a Test Case for a Template Version and is in the process of configuring it, filling out all of the fields necessary to hydrate the document for rendering it based on the provided input. While in this state the Changeset Editor or Changeset Publisher can run the Test Cases and view the rendered PDF results, but they can not move those Test Cases to PASSED. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TestCaseDeletedEvent</td><td>DELETED</td><td>Test Case is deemed no longer needed and is DELETED by a user.</td></tr><tr><td>TestCaseCompletedEvent</td><td>READY FOR REVIEW</td><td>Test Case has been successfully run and the results are READY FOR REVIEW.</td></tr></table>			EVENT	TO	DESCRIPTION	TestCaseDeletedEvent	DELETED	Test Case is deemed no longer needed and is DELETED by a user.	TestCaseCompletedEvent	READY FOR REVIEW	Test Case has been successfully run and the results are READY FOR REVIEW.
EVENT	TO	DESCRIPTION										
TestCaseDeletedEvent	DELETED	Test Case is deemed no longer needed and is DELETED by a user.										
TestCaseCompletedEvent	READY FOR REVIEW	Test Case has been successfully run and the results are READY FOR REVIEW.										
READY FOR REVIEW	A Test Case has been successfully run, and the rendered PDF results can be reviewed and marked as either FAILED or PASSED. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TestCaseFailedEvent</td><td>FAILED</td><td>The Test Case has been run and the user was dissatisfied with the results, and so has marked it as FAILED.</td></tr><tr><td>TestCasePassedEvent</td><td>PASSED</td><td>The Test Case has been run and the user was satisfied with the results, and so has marked it as PASSED.</td></tr></table>			EVENT	TO	DESCRIPTION	TestCaseFailedEvent	FAILED	The Test Case has been run and the user was dissatisfied with the results, and so has marked it as FAILED.	TestCasePassedEvent	PASSED	The Test Case has been run and the user was satisfied with the results, and so has marked it as PASSED.
EVENT	TO	DESCRIPTION										
TestCaseFailedEvent	FAILED	The Test Case has been run and the user was dissatisfied with the results, and so has marked it as FAILED.										
TestCasePassedEvent	PASSED	The Test Case has been run and the user was satisfied with the results, and so has marked it as PASSED.										
FAILED	After evaluation by a Changeset Editor or Changeset Publisher, the Test Case has been deemed unacceptable. From here, the Test Case can be recalled back to DRAFT, by either the Changeset Editor or Changeset Publisher, for further modification. Transitions from this state: <table><tr><th>EVENT</th><th>TO</th><th>DESCRIPTION</th></tr><tr><td>TestCaseRecalledEvent</td><td>DRAFT</td><td>The Changeset Editor or Changeset Publisher recalled the Failed Test Case back to DRAFT to make changes as needed.</td></tr></table>			EVENT	TO	DESCRIPTION	TestCaseRecalledEvent	DRAFT	The Changeset Editor or Changeset Publisher recalled the Failed Test Case back to DRAFT to make changes as needed.			
EVENT	TO	DESCRIPTION										
TestCaseRecalledEvent	DRAFT	The Changeset Editor or Changeset Publisher recalled the Failed Test Case back to DRAFT to make changes as needed.										
PASSED	After evaluation by a Changeset Editor or Changeset Publisher, the Test Case has been deemed acceptable and has been marked as PASSED.											
DELETED	This Test Case is no longer needed and has been DELETED.											

Letter Instance Lifecycle



DRAFT	This Letter Instance has been instantiated based on an ACTIVE Template Version. Upon entering and exiting from this state, the ECM-L system will pull in everything needed from Resources configured in the Template Version. In the Letter Manager, when a Letter Editor is creating a new Letter Instance, it will always be instantiated based on the most recent ACTIVE Template Version. System Actors are able to generate Letter Instances based on older ACTIVE Template Versions so that they have access to a reliable API data model. Note: • <i>DRAFT Letter Instances can be rendered as a PDF but will be contain a "DRAFT" watermark.</i> Transitions from this state:		
	EVENT	TO	DESCRIPTION
	LetterInstanceDeletedEvent	DELETED	Letter Instance no longer needed and has been DELETED.
	LetterInstanceDraftCompleteEvent	READY FOR REVIEW	Letter Instance based on ACTIVE Template Version created and marked READY FOR REVIEW.

READY FOR REVIEW	This Letter Instance contains all elements required by the Template Version (e.g., inputs, attachments, inclusions, etc.) and has been submitted for review.		
	<i>Note:</i> - For some Lines of Business a single user will move the Letter Instance all the way through the lifecycle to Finalization without a separate user approving the Letter Instance. - READY FOR REVIEW Letter Instances can be rendered as a PDF but will contain a "READY FOR REVIEW" watermark.		
	Transitions from this state:		
	EVENT	TO	DESCRIPTION
	LetterInstanceDraftRejectedEvent	DRAFT	Changes need to be made to the Letter Instance before it can pass review and start the finalization process, so a Letter Finalizer has returned the Letter Instance to DRAFT for further modification.
	LetterInstanceRecreatedforNewVersionEvent	DRAFT	This Letter Instance has been re-instantiated against a newer Version of the Template. This is optional after a Changeset Publisher has published a newer Version of the Template.
FINALIZED	LetterInstanceFinalizedEvent	FINALIZED	The Letter Instance has been reviewed, and the resulting rendered PDF has been deemed acceptable.
	LetterInstanceDeletedEvent	DELETED	Letter Instance is no longer needed and has been DELETED.
FINALIZED	This Letter Instance has been approved and made Final. In this state the Letter Instance is immutable and will persist unchanged for historical purposes, and every subsequent rendering of the Letter Instance will have the same result for all time. <i>Note:</i> FINALIZED Letter Instances can be rendered as a PDF with no watermark.		
DELETED	This Letter Instance is no longer needed and has been DELETED. <i>Note:</i> DELETED Letter Instances can no longer be rendered.		
ARCHIVED	After a given period of time, the Letter Instance will be ARCHIVED as per applicable archival procedure.		

Data Models

Conceptual Data Models (DIV-1)

The following diagrams capture high level data concepts of ECM-L.

Changeset Locking

Changeset states are categorized as either Unfinished or Finished. A Changeset is "Finished" when it is Published or Canceled; a Changeset in any other state is "Unfinished" (i.e., Draft, Ready to Test, Ready for Approval, Approved, or Scheduled). When a Changeset is in an Unfinished state, the Templates in that Changeset are essentially "Locked" and can only exist in that single Unfinished Changeset and cannot be included in any other Changeset.

Templates in an "Unfinished" Changeset are essentially "Locked" and cannot be used in other Changesets.

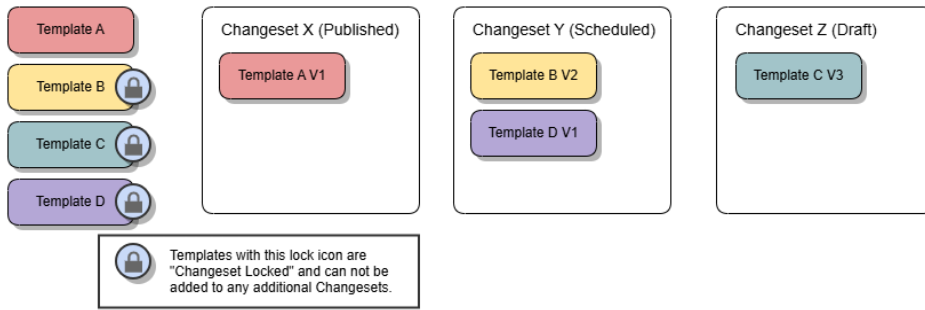
"Unfinished" Changeset States:

- Draft
- Ready to Test
- Ready for Approval
- Approved
- Scheduled

After a Changeset is moved to a "Finished" state the Templates that it has "Locked" become "Unlocked" and can now be used in other Changesets again.

"Finished" Changeset States:

- Canceled
- Published

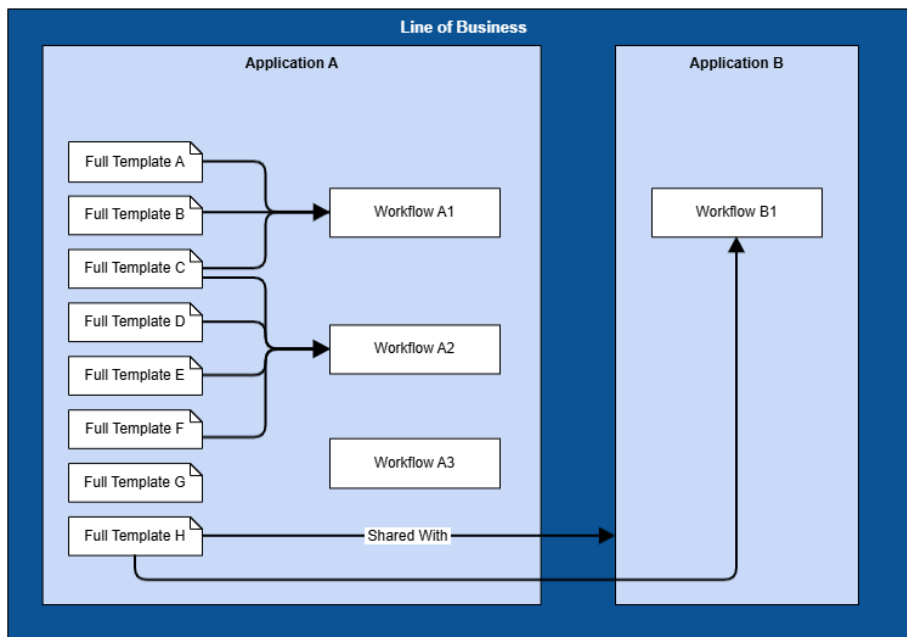


Application, Workflow and Full Template Alignment

This diagram illustrates the relationship between workflows, and templates within an application under the line of business. It consists of three main components: Applications, Workflows and Full Templates.

- **Applications:** Applications are what own a Template. LOB Admins can share a Template owned by one Application with another Application.
- **Full Templates:** Full Templates are user created Line of Business tailored Letter templates for a type of correspondence.
- **Workflows:** Workflows represent discrete tasks and sub processes within which end users of an Application may interact with the ECM-L.

By aligning Full Templates to a Workflow, we are indicating to the system that a user is able to generate Letter Instances based on those Full Templates when a user accesses that Workflow in the Letter Manager.



The diagram above shows how Templates relate to Workflows.

Full Templates A, B, and C, are aligned to Workflow A1, so a user accessing the Letter Manager through that Workflow will be able to choose from those Templates when creating a Letter Instance. Even though Full Template D is a part of the same Application, since it is not aligned to Workflow A1, it will not appear in the list of available Templates for that Workflow.

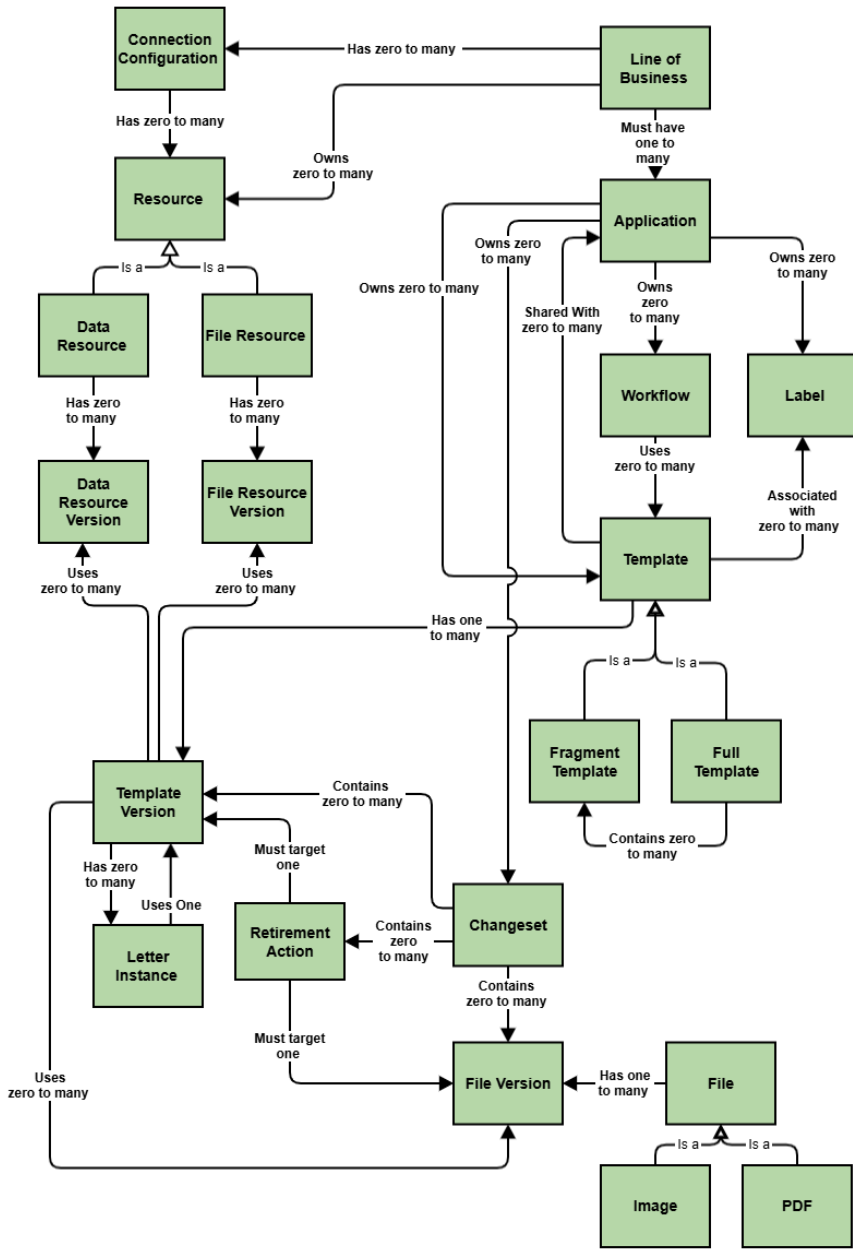
Full Template H, which belongs to Application A, is *shared* with Application B. Since it's been shared with Application B, the Application B users are able to align that Template to Workflow B1. Now in the Letter Manager, users within Workflow B1, in Application B, are able to create a Letter Instance using Full Template H.

Application	Workflow	Available Templates
A	A1	A, B, C

B	A2	C, D, E, F
	A3	None
	No Workflow	All
	B1	H
	No Workflow	None

Key Terms and Concepts

This is a diagram illustrating the basic high level concepts and how they relate to one another.



Key Term	Description
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Application	The name of a product or service within a Line of Business that integrates with and interacts with the ECM-L components.
Changeset	A collection of Versioned changes that a user wants to make to Templates and/or Files, including new Template Versions, File Versions, and/or Retirement Actions. This is the mechanism for managing the content within the Template Manager, including creating, testing and publishing new Template Versions, as well as retiring Template Versions.
Connection Configuration	A collection of the Configuration information necessary for the ECM-L components to access another system's API so that it's Resources can be used by Templates and Letters.
Data Resource	Structured data resulting from a successful GET operation on an API.
File	A File that can be used by a Template such as a PDF or an Image. These live within the ECM-L eco-system and are managed within Changesets in the Template Manager like a Template, as opposed to File Resources which point to Files hosted externally.
File Resource	A type of Resource used to pull in different file objects for use in a Template, such as an Image or PDF.
File Version	A specific version of a File within the ECM-L.
Fragment Template	A Template that represents a piece of content that can be included inside of another Template. A Fragment Template cannot be instantiated as a Letter Instance on its own.
Full Template	A Template that represents the full content of a piece of correspondence, possibly including Fragment Templates within it.
Image	An Image is a type of File that can be inserted into Template content and rendered with a Letter Instance as a part of the PDF.
Label	Used by Changeset Editors to categorize and organize Templates.
Letter Instance	The instantiation of a Template Version. A Letter Instance can be provided in different forms, such as JSON data, or as a PDF.
Line of Business	An organizational segment of operational and strategic activities. All Applications belong to a single Line of Business.
PDF	Portable Document File, a type of File that can be used by a Template as an Attachment.
Resource	Data or a File that is provided by an external system outside of the ECM-L eco-system, via one of the API's configured through a Connection Configuration.
Retirement Action	Retirement Actions point to versioned content such as a File Version or a Template Version and upon publication will retire that specific version of that content. (See Retirement)
Template	A versioned representation of a type of document, potentially containing reusable content such as Fragment Templates and contextual data provided by Data Resource placeholders, necessary to successfully instantiate and render a Letter Instance. Templates belong to a specific Application, but can be shared between Applications. There are two types of Templates - Full Templates and Fragment Templates.
Template Version	A specific version of a Template.
Workflow	The LoB-specific business process context in which the Letter Manager MFE operates by organizing and categorizing full templates and associating specific Key Parameters required for letter generation within an application.

Retirement

In the ECM-L ecosystem *Retirement* refers to the process of making an item (such as a Template Version, Key Parameter, Workflow) unavailable for future use. Once an item is *Retired*, it no longer appears for selection in any lists that would include it as an option, and no new items can be created using that item. However, any existing configurations or places that already made use of the retired item will continue to function just as before. This ensures safety and continuity for current configurations while preventing the use of outdated or deprecated options moving forward.

Item	Retirement Impact
File Version	New Template Versions can no longer be configured to use this File Version - it is no longer available for use inside of the content of a Template in the case of Images, or as an attachment in the case of PDF's. Existing Template Versions already configured with this File Version can continue to use it as before.

Key Parameter	New Resource Versions and Workflows can no longer be configured to use this Key Parameter - it does not appear in the list of available Key Parameters. Existing Resources and Workflows already configured with this Key Parameter can continue to use it as before.
Resource Version	New Template Versions can no longer be configured to use this Resource Version - it does not appear in the list of available Resources. Existing Template Versions already configured with this Resource Version can continue to use it as before.
Template Version	New Letter Instances can no longer be instantiated using this Template Version. Existing Letter Instances instantiated from this Template prior to its Retirement can still be worked and viewed as before.
Workflow	New Letter Instances can no longer be created within this Workflow. Existing Letter Instances created before the Workflow was Retired can still be worked and viewed as before.

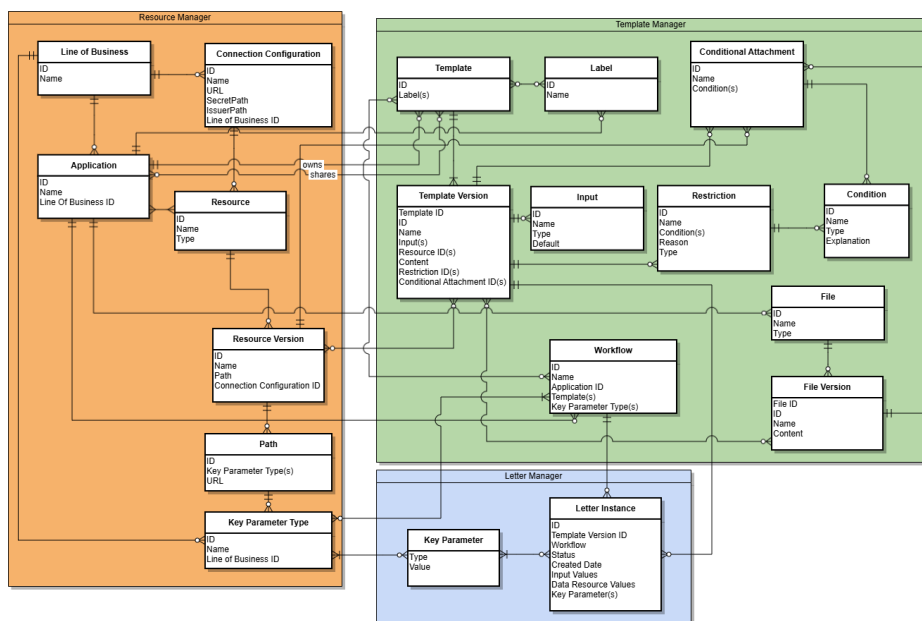
Template Version Duplication and Restoring Retired Template Versions

Users in the Template Manager are able, when creating a new Template Version, to optionally choose to make that new Template Version a copy of an existing Template Version. The chosen source Template Version, that the user wishes to copy, can exist in any state. If a user wants to restore a Template Version they regret retiring, they can create a new Version of that Template using the retired Version as a source, effectively creating a new Template Version that mirrors the content and the configuration options of the retired Version.

Some configuration options, however, may not be able to be copied, such as the inclusion of a Fragment Template or a Resource that has been retired. Since this duplicate is a new Template Version, it is bound by the same restrictions as any other new Template Version.

Entity Relationship Diagram

This Entity Relationship Diagram illustrates how the high level concepts of the ECM-L systems interact with and relate to one another.



Resource Manager

Line of Business

A collection of Applications / services serving a specific customer need or business function. Applications, Resources, Key Parameter Types, and Connection Configurations all belong to a specific Line of Business.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.

Application

The product or service that belongs to a Line of Business which needs to interact with the ECM-L System.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Line of Business ID	The UUID of the Line of Business to which it belongs.

Connection Configuration

All of the configuration information necessary for the ECM-L to successfully interact and integrate with a specific external API. These are loaded into the Resource Manager via a configuration file and then are made available for use by Resources via the Resource Manager.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
URL	The environment-specific URL for the API.
Secret Path	The path to the secret in Vault.
Issuer Path	The path to the issuer in Vault.
Line of Business ID	The UUID of the Line of Business to which it belongs.

Resource

The information necessary to gather content and data from outside of the ECM-L system that can be used within the documents that it creates and renders.

Attribute Name	Description
ID	Its immutable UUID.
Name	The user friendly name of a Resource.
Type	The type of the Resource.

Resource Version

A specific Version of a Resource.

Attribute Name	Description
ID	Its immutable UUID.
Paths	A list of paths that can be used to access the Resource, depending on the given Key Parameter Type.
Connection Configuration ID	The ID of the Connection Configuration it uses.

Path

The information related to a specific operation / API path that will allow the ECM-L System to access the information in a Resource Version.

Attribute Name	Description
ID	Its immutable UUID.
Key Parameter Type(s)	The IDs of the Key Parameters required for this path.
URL	The path to the endpoint in the API represented by the chosen Connection Configuration.

Key Parameter Type

A Line of Business and Application specific data type used to organize and contextualize correspondence and used as a Parameter to gather contextual data from a Resource.

Attribute Name	Description
----------------	-------------

ID	Its immutable UUID.
Name	Its user-friendly name.
Line of Business ID	The UUID of the Line of Business to which it belongs.

Template [Manager](#)

Workflow

The Workflow or sub process in which Letters are created within a given Application.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Application ID	The UUID of the Application to which it belongs.
Templates	The UUIDs of any Templates associated with this Workflow.
Key Parameter Type(s)	The type(s) of Key Parameter given for the context.

Template Label

A label is used to help users assign meaning and organizational order to their Templates.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.

File

A piece of file-based content. These are versioned and each File Version is managed by a Changeset, similar to Templates. There are two different types of Files - PDF's and Images

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Type	The type of the File - either a PDF or an Image.

File Version

The versioned File content.

Attribute Name	Description
File ID	The immutable UUID of the File this is Versioning.
ID	Its immutable UUID.
Name	Its user-friendly name.
Content	The actual content of the File.

Template

A collection of versioned configuration information and content required to manage a specific type of correspondence. There are two different types of Templates - Full Templates and Fragment Templates.

Attribute Name	Description
ID	Its immutable UUID.

Labels	The labels to which this Template has been aligned.
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Template Version

The versioned configuration information and content required to manage a specific type of correspondence.

Attribute Name	Description
Template ID	The immutable UUID of the Template this is Versioning.
ID	Its immutable UUID.
Name	Its user-friendly name.
Inputs	A collection of inputs that can be used to hydrate the content of the Template with user created data.
Resource IDs	The UUID's of any Resources that are aligned to this Template, that can be used to hydrate the content of the Template with contextual data.
Content	The actual FreeMarker and textual content of the Template.
Restriction IDs	The UUID's of any restrictions that should be applied to correspondence created based on this Template.
Conditional Attachment IDs	The UUID's of any attachments that should be conditionally added to correspondence created based on this Template.

Conditional Attachment

Files like PDF's that are attached to correspondence if certain conditions are met.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Conditions	A list of conditions that once met will result in it being attached to the correspondence.

Condition

A comparison or check used in Restrictions and Conditional Attachments.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Type	The condition type, such as a string-match or a boolean condition.
Explanation	A user friendly explanation that describes the condition in plain English.

Restriction

A condition based restriction on creating correspondence related to a specific Template Version.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Conditions	A list of the UUIDs for conditions that if met result in a Template Version being restricted.
Reason	A user-friendly description of the reason for this restriction.
Type	Determines the impact of this restriction - such as hiding it from the list of available Templates, or disabling it and indicating why using the given Reason.

Input

A way to collect data from a user that can be used to hydrate the content of a Template.

Attribute Name	Description
ID	Its immutable UUID.
Name	Its user-friendly name.
Type	Determines the type of input, which informs things like the formatting options in the Content Editor of the Template Manager and which form elements to present to the user in the Enterprise Letter Virtual Interview Screen (ELVIS).
Default	The optional default value of the Input.

[Letter Manager](#)

Letter Instance

An instantiated Letter based on the content and configuration options outlined in a specific version of a Full Template.

Attribute Name	Description
ID	Its immutable UUID.
Template Version ID	The UUID of the Template Version for which this instance was created and will be rendered against.
Workflow	The Workflow in which this Letter Instance was created.
Created Date	The date this instance was created.
Input Values	A collection of the data collected from users based on the Inputs defined in the Template Version.
Data Resource Values	A collection of the data collected from data resources based on the Resources defined in the Template Version.

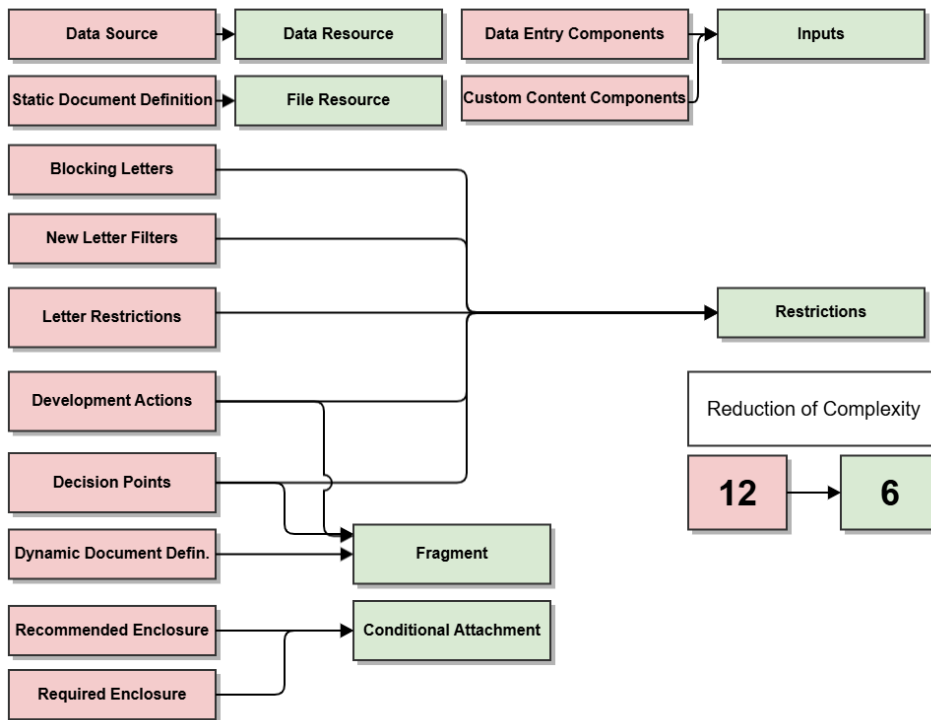
Key Parameter

A type / value pair that informs us of the context in which a Letter Instance was created.

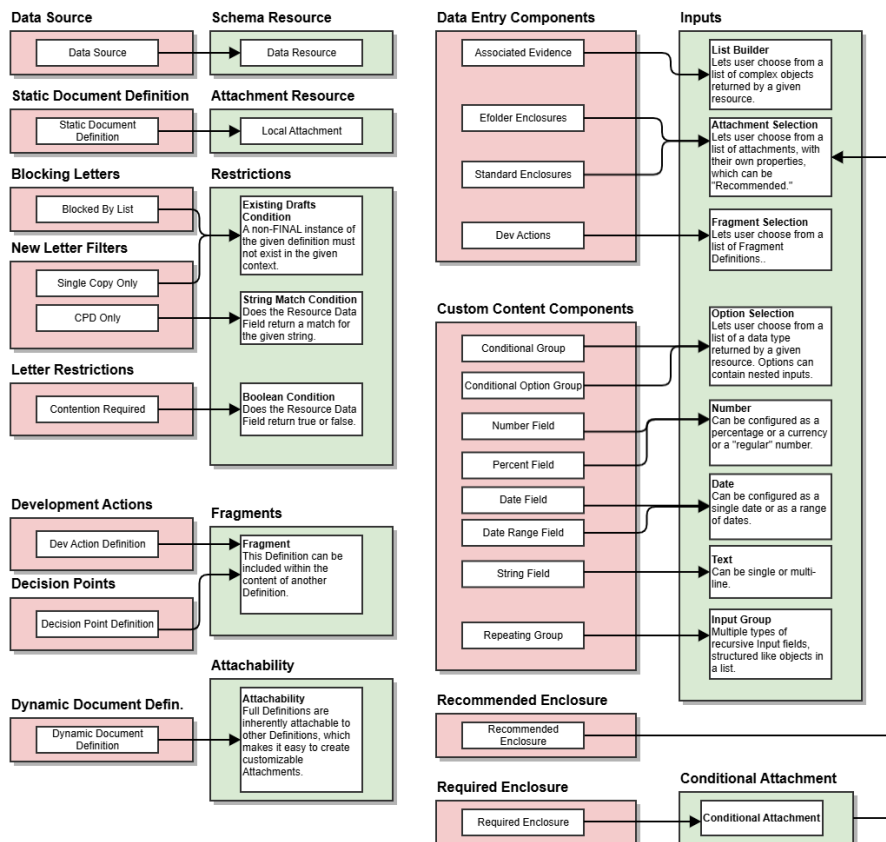
Attribute Name	Description
Type	The ID of the Key Parameter Type that relates to the given value.
Value	The value provided either automatically by the Line of Business system or manually entered by an end user.

Logical Data Viewpoint Model (DIV-2)

The following diagram defines relationships between data elements, providing a structured representation of data flows, entities, and attributes.



Lower Level Rosetta Stone



Physical Data Viewpoint Model (DIV-3)

N/A

Process View (Business Process Model)

The Process View describes the behavior of the systems, the sequence of events, and how various tasks combine to the functionality described in the Use-Case View. It decomposes the systems into lightweight processes (single threads of control) and heavyweight processes (groupings of lightweight processes). The Process View describes how systems interact, detailing the timing of messages or events.

Main Business Process Model (OV-6d)

Intake Processes

The following diagrams delineate the key processes and workflows within ECM-L, mapping the sequence of activities and information flow that drive operations.

Line of Business Intake Process

The intake process for a new Line of Business involves an ECM-Line of Business Intake Form, which outlines how the new Line of Business interacts with ECM-L system and its components, and identifies the Application(s) and any Connection Configurations the Line of Business wants to include in the intake process. The Line of Business will submit the Line of Business Intake Form, and Connection Configuration Intake Form if needed, and create a Jira ticket. After submission of the Intake Form(s), there will be an Integration Meeting with stakeholders to introduce ECM-L capabilities, integration points and connection configurations.

[Enterprise Correspondence Management - Line of Business Intake Form](#)

Application Intake Process

The intake process for a new Application is performed by configuration outside of the ECM-L user interface. It involves Service Tickets and System Teams and successful processing results in an Application becoming usable in ECM-L.

[Enterprise Correspondence Management - Application Intake Form](#)

Connection Configuration Intake Process

The connection configuration intake process results in a successful integration between the ECM-L applications and an external Line of Business API. This process begins with a Connection Configuration Intake Form filled out by the Line of Business, which results in a Change Request that loads the necessary Secret / Issuer key pair into the external API and into the individual ECM-L components, and ultimately ends with a Connection Configuration in the Provisioned status, ready to be used.

The Connection Configuration intake form for a new LoB requesting connection to ECM-L will require the following:

- Point of Contact
- Line of Business
- Connection Name
- API Registration
- Vault Location
- OAS URL
- Certificates and Secrets
- API Permissions and Policies
- Swagger or OAS Specification
- Input Parameters
- Security Needs
- Transport Layers
- JWT
- JSON Path

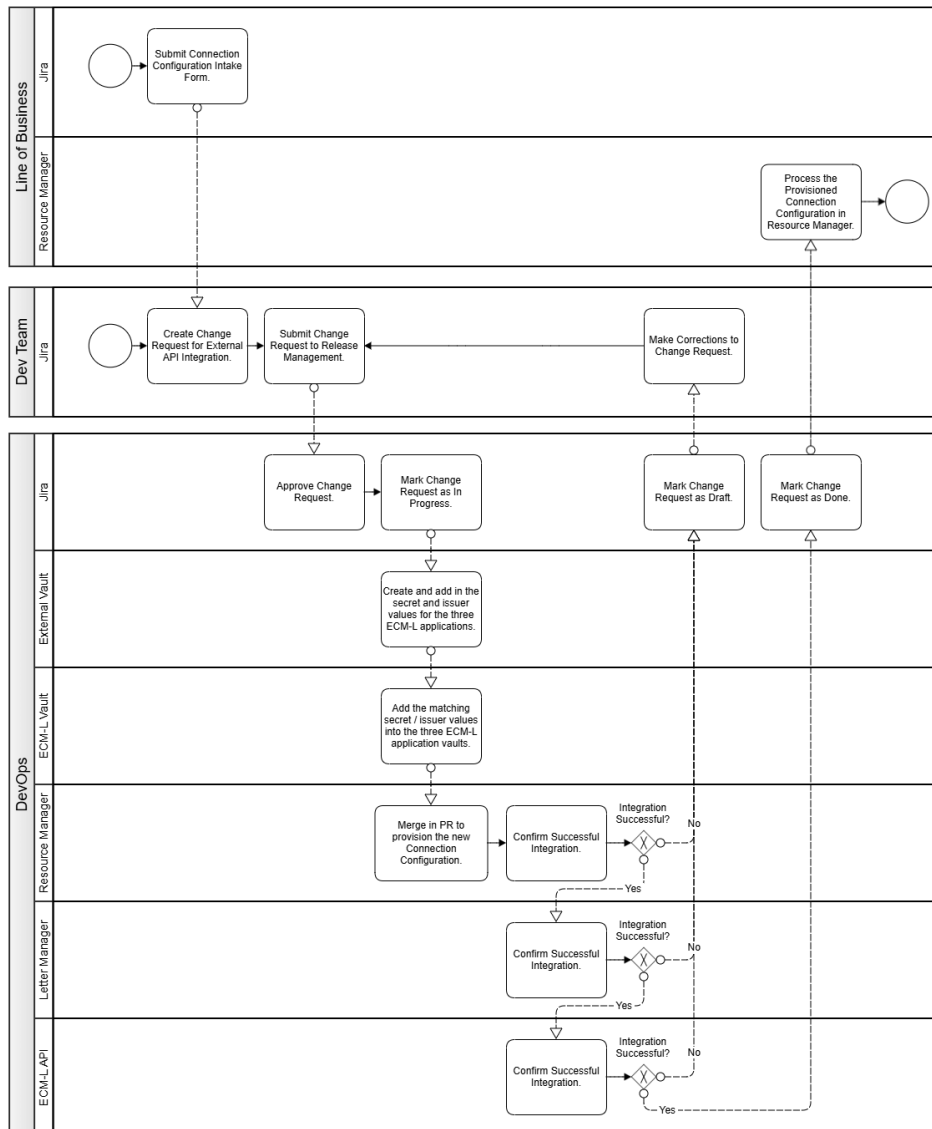
Here is an example Change Request ticket that can be cloned in Jira:

[BPS-66120 - Getting issue details...](#)

STATUS

Process Workflow for Connection Configuration Intake

This process guides the intake and integration of new connections. It includes submission, approval, configuration, and confirmation across multiple systems.

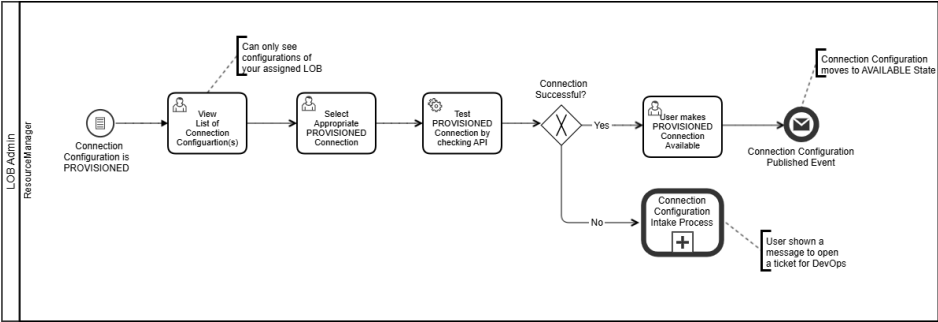


Task	Description
Submit Connection Configuration Intake Form	Line of Business submits an intake form to request a new connection configuration.
Create Change Request for External API Integration	Dev Team creates a change request to initiate the API integration process.
Submit Change Request to Release Management	Dev Team submits the change request for formal approval through release management.
Approve Change Request	Approval is given to proceed with the connection changes.
Mark Change Request as In Progress	The change request status is updated to indicate active work.
Create and Add Secret/Issuer Values in ECM-L Vault and External Vault	DevOps team populates secret and issuer values needed for integration in ECM-L Vault and External Vault.
Merge PR to Provision Connection Configuration	Pull request is merged to apply the new configuration.
Confirm Successful Integration (Resource Manager)	Confirm that the integration is successful in the Resource Manager.
Confirm Successful Integration (Letter Manager)	Verify that Letter Manager has been successfully integrated.
Confirm Successful Integration (ECM-L API)	Ensure that ECM-L API reports successful integration.
Mark Change Request as Done	Dev Team marks the request as completed once all validations pass.

Process the Provisioned Connection Configuration in Resource Manager	Resource Manager finalizes configuration.
Make Corrections to Change Request	If issues arise, the Dev Team adjusts the change request and may re-submit it.
Mark Change Request as Draft	If changes are incomplete, the request is returned to draft status for revision.

Make Connection Available

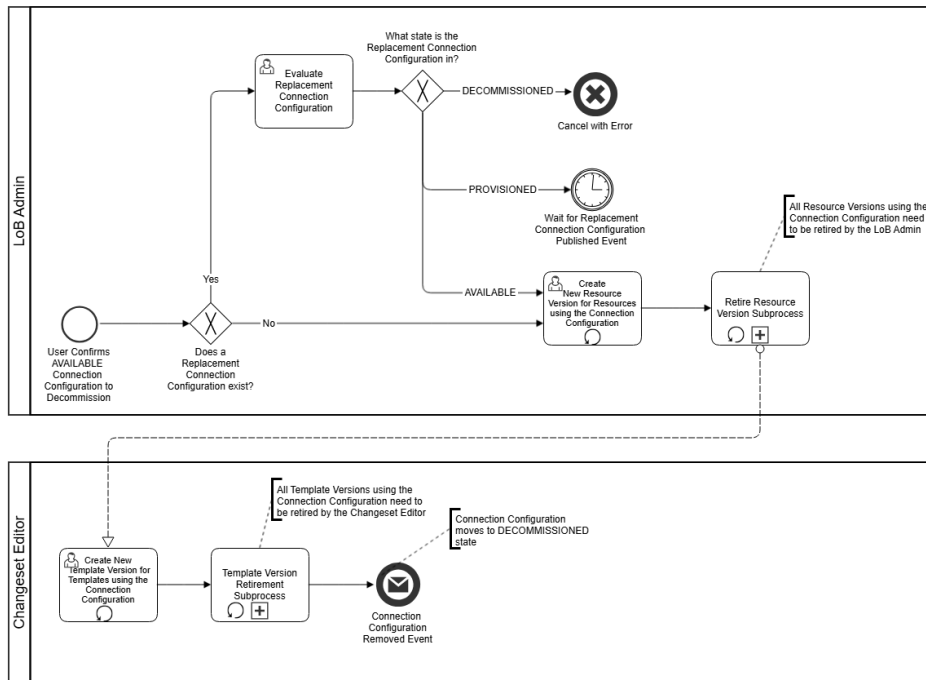
The Line of Business Administrator is able to make a Connection Configuration available.



Task	Description
Connection Configuration is PROVISIONED	The process begins when a Connection Configuration moves to PROVISIONED status.
View List of Connection Configuration(s)	The LOB Admin views the list of Connection Configurations assigned to their Line of Business (LOB).
Select Appropriate PROVISIONED Connection	The LOB Admin selects a PROVISIONED Connection Configuration to make AVAILABLE.
Test PROVISIONED Connection by checking API	The LOB Admin tests the connection's API to ensure it works.
Connection Successful?	Decision point: Was the API connection test successful?
If Yes Select Appropriate PROVISIONED Connection	If the test is successful, the LOB Admin makes the PROVISIONED Connection Configuration Available.
Connection Configuration Published Event	Connection Configuration Published Event is triggered <i>*Connection Configuration moves to AVAILABLE state</i>
If No Connection Configuration Intake Process	If the test fails, the LOB Admin must return to the Connection Configuration Intake Process .

Connection Configuration Decommissioning Process

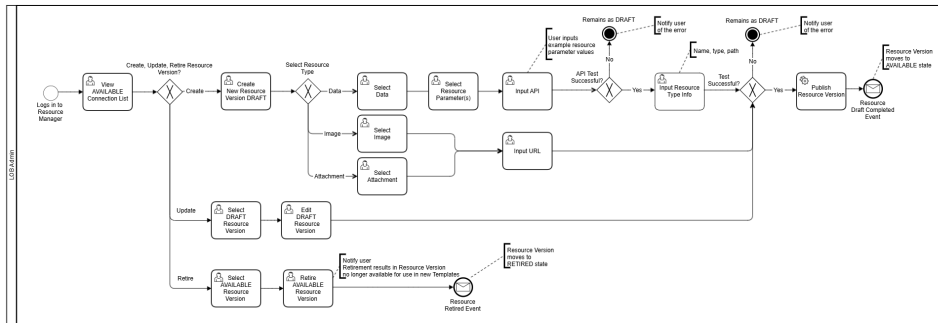
The process of Decommissioning a Connection Configuration is first initiated through Jira. The flow below handles Retiring a Connection Configuration while considering replacements and dependencies. It ensures Resource and Template clean-up before removal.



Task	Description
User Confirms AVAILABLE Connection Configuration to Decommission	The LoB Admin confirms the AVAILABLE Connection Configuration to Decommission.
Does a Replacement Connection Configuration Exist?	Decision point: Does a replacement for the Connection Configuration already exist?
If No Create New Resource Version for Resources using the Connection Configuration	If a replacement Connection Configuration does not exist, the LoB Admin creates new Resource Version (s) for Resources using the Connection Configuration.
If Yes Evaluate the Replacement Connection Configuration	If a replacement Connection Configuration exists, that Connection Configuration must be evaluated for readiness and availability.
What state is the Replacement Connection Configuration in?	Decision point: What state is the Replacement Connection Configuration in?
If DECOMMISSIONED Cancel with Error	If the state of the replacement Connection Configuration is DECOMMISSIONED, the process is canceled due to error because a DECOMMISSIONED Connection Configuration cannot be used as replacement.
If PROVISIONED Wait for Replacement Connection Configuration Published Event	If the state of the replacement Connection Configuration is PROVISIONED, pause the process until the replacement Connection Configuration Published Event is triggered (putting the Connection Configuration in AVAILABLE)
If AVAILABLE Create New Resource Version for Resources using the Connection Configuration	If the state of the replacement Connection Configuration is AVAILABLE, the LoB Admin creates new Resource Version(s) for resources using the Connection Configuration.
Retire Resource Version Subprocess	The LoB Admin begins the Retire Resource Version Subprocess (NEED TO CREATE THIS SUBPROCESS) *Note: all Resource Versions using the Connection Configuration need to be retired
Create New Template Version for Templates using the Connection Configuration	The Changeset Editor creates new Template Version(s) for Template using the Connection Configuration.
Template Version Retirement Subprocess	The Changeset Editor begins the Template Version Retirement Subprocess . *Note: all Template Versions using the Connection Configuration need to be retired
Connection Configuration Removed Event	Connection Configuration Removed Event is triggered. *Connection Configuration moves to DECOMMISSIONED state

Create / Update / Retire Resource Version

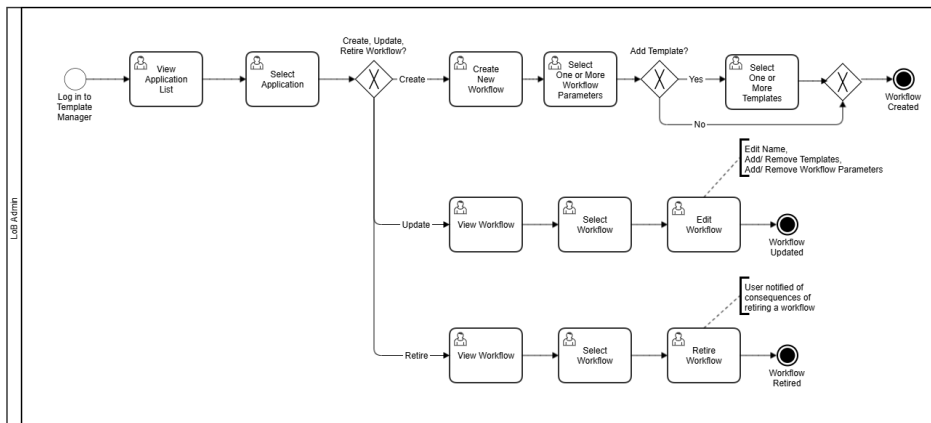
The model demonstrates the creation, modification and retirement of a Resource Version within the Resource Manager. Three types of Resources can be added, each requiring testing before they become available for use.



Task	Description
Log In to Resource Manager	The process begins when the LOB Admin logs into Resource Manager.
View AVAILABLE Connection List	The LOB Admin sees a list of AVAILABLE Connection Configuration(s).
Create, Update or Retire Resource Version?	Decision point: Do you want to Create, Update or Retire a Resource Version?
If Update Select DRAFT Resource Version	If the Resource Version is already created, the LOB Admin selects the DRAFT Resource Version.
Edit DRAFT Resource Version	The LOB Admin edits the DRAFT Resource Version. The Resource Version will then need to be tested prior to making AVAILABLE.
If Create Create New Resource Version DRAFT	The LOB Admin initiates the creation of a new Resource Version, which is initially in a DRAFT state.
Select Resource Type?	Decision point: the LOB admin chooses the type of Resource to create: Data, Image, or Attachment?
If Data Select Data	If Data, the LOB Admin proceeds to select the specific Data Resource.
Select Resource Parameter(s)	The LOB Admin selects the Resource Parameter(s).
Input API	The LOB Admin provides API information needed to test, such as example Resource Parameter Values.
API Test Successful?	Decision point: The systems tests the provided API connection for validity.
If No Remains as DRAFT	If the test was unsuccessful, the LOB Admin will be notified of the error and the Resource will remain in DRAFT status.
If Yes Input Resource Type Info	If the test was successful, the LOB Admin enters the Resource Type information, such as name, type and path for the Resource.
If Image Select Image	If Image, the LOB Admin proceeds to select the specific Image Resource.
If Attachment Select Attachment	If Attachment, the LOB Admin proceeds to select the specific Attachment Resource.
Input URL	For both "Image" and "Attachment", the LOB Admin provides the URL information needed.
Test Successful?	Decision point: The system tests if the information is valid for the given Resource.
If No Remains as DRAFT	If the test was unsuccessful, the LOB Admin will be notified of the error and the Resource will remain in DRAFT status.
If Yes Publish Resource Version	If the test was successful, the Resource Version is published by the system.
Resource Draft Completed Event	Resource Draft Completed Event is triggered. <i>*Resource Version moves to AVAILABLE state</i>
If Retire Select AVAILABLE Resource Version	The LOB Admin selects a Resource Version in the AVAILABLE state that they would like to Retire.
Retire AVAILABLE Resource Version	The LOB Admin retires the AVAILABLE Resource Version and is notified of the consequence of Retiring.
Resource Retired Event	Resource Retired Event is triggered. <i>*Resource Version moves to RETIRED state</i>

Create / Update / Retire Workflow

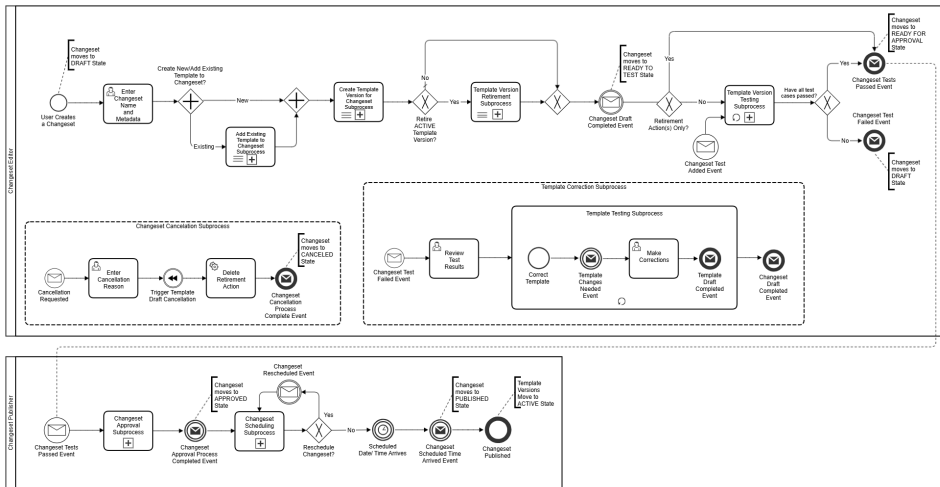
This model outlines the steps the LoB Admin follows to create, update, or retire a Workflow.



Task	Description
Log in to Template Manager	The LoB Admin logs into Template Manager.
View Application List	The LoB Admin sees the list of available Applications.
Select Application	The LoB Admin selects a specific Application to create a Workflow for.
Create, Update, Retire Workflow?	Decision point: do you want to create, update, or retire a Workflow?
If Create Create New Workflow	If create, the LoB Admin selects create new Workflow.
Select One or More Workflow Key Parameters	The LoB Admin selects one or more Workflow Key Parameters.
Add Template?	Decision point: do you want to add Template(s) to your Workflow?
If Yes Select One or More Templates	If the LoB Admin wants to add Template(s), they will select or or more Templates to add.
Workflow Created	The Workflow is created in the system.
If No Workflow Created	If the LoB Admin does not want to add Template(s), the Workflow is still created in the system.
If Update View Workflow	The LoB Admin views the list of Workflows under the selected application.
Select Workflow	The LoB Admin selects a specific Workflow to update.
Edit Workflow	The LoB Admin edits the Workflow by changing its name, adding/removing Template(s), and/or add/removing Workflow Parameters.
Workflow Update	The system updates the Workflow.
If Retire View Workflow	The LoB Admin view the list of Workflows under the selected Application.
Select Workflow	The LoB Admin selects the Workflow they want to retire.
Retire Workflow	The LoB Admin initiates the retirement process. A notification alerts them of the consequences of this action.
Workflow Retired	The system retires the selected Workflow.

Overall Changeset Process (**awaiting final approval)

This model outlines the complete lifecycle of a Changeset from creation to publishing. It illustrates key states, decision points, and transitions across testing, approval, and scheduling.



Overall Changeset Process

Task	Description
User Creates a Changeset	The Changeset Editor creates a Changeset in Template Manager. <i>*Changeset moves to DRAFT state</i>
Enter Changeset Name and Metadata	The Changeset Editor enters the Changeset's name and metadata. Metadata may includes notes as to why the Changeset is being created.
Create New/ Add Existing Template to Changeset	Decision point: Do you want to create a new Template or add an existing Template to the Changeset?
If Add Existing Add Existing Template to Changeset Subprocess	If the Changeset Editor wants to add an existing Template to the Changeset, they will follow the Add Existing Template to Changeset Subprocess .
Create Template Version for Changeset Subprocess	Once the Changeset Editor has added an existing Template OR if the Changeset Editor wants to create a new Template, they will follow the Create Template Version for Changeset Subprocess .
Retire ACTIVE Template Version?	Decision point: Do you want to retire an ACTIVE Template Version?
If Yes Template Version Retirement Subprocess	If the Changeset Editor wants to retire an ACTIVE Template Version, they will follow the Template Version Retirement Subprocess .
Changeset Draft Completed Event	Changeset Draft Completed Event is triggered. <i>*Changeset moves to READY TO TEST state</i>
Retirement Action(s) Only?	Decision point: Is this Changeset performing Retirement Action(s) only?
If Yes Changeset Tests Passed Event	If the Changeset was created to perform only Retirement Action(s), the Changeset does NOT need to go through testing and the Changeset Tests Passed Event is triggered. <i>*Changeset moves to READY FOR APPROVAL state</i>
If No Template Version Testing Subprocess	If the Changeset is not Retirement Action(s) only, it will need to go through the Template Version Testing Subprocess .
Have all test cases passed?	Decision point: Have all the Test Cases PASSED?
If Yes Changeset Tests Passed Event	If all of the Test Cases have PASSED, the Changeset Tests Passed Event is triggered. <i>*Changeset moves to READY FOR APPROVAL state</i>
If No Changeset Test Failed Event	If any of the Test Cases FAIL, the Changeset Test Failed Event is triggered. <i>*Changeset moves back to DRAFT state</i>
Changeset Tests Passed Event	The Changeset Publisher's process starts once the Changeset Tests Passed Event is triggered and the Changeset is in READY FOR APPROVAL.
Changeset Approval Subprocess	The Changeset Publisher will Approve Changesets through the Changeset Approval Subprocess .

Changeset Approval Process Completed Event	Changeset Approval Process Completed Event is triggered, <i>*Changeset moves to APPROVED state</i>
Changeset Scheduling Subprocess	Once the Changeset is APPROVED, the Changeset Publisher is able to Schedule the Changeset through the Changeset Scheduling Subprocess .
Reschedule Changeset?	Decision point: Do you need to reschedule the Changeset?
If Yes Changeset Rescheduled Event	If you need to reschedule the Changeset, a Changeset Rescheduled Event is triggered.
Changeset Scheduling Subprocess	The Changeset Rescheduled Event will trigger the Changeset Publisher to go back to the Changeset Scheduling Subprocess to reschedule the Changeset.
If No Scheduled Date/Time Arrives	If you do not need to reschedule or you do not reschedule by the time the Scheduled date/time arrives, the Changeset Scheduled Time Arrived Event is triggered.
Changeset Scheduled Time Arrived Event	Once the Scheduled date/time arrives, the Changeset will be PUBLISHED. <i>*Changeset moves to PUBLISHED state</i>
Changeset Published	<i>*Template Versions move to ACTIVE state</i>

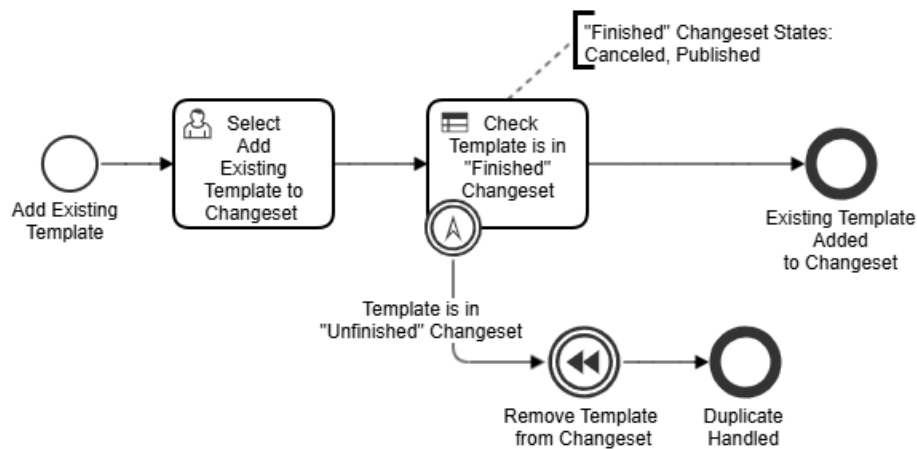
Changeset Cancellation Subprocess

Task	Description
Cancellation Request	The process begins when a cancellation request has been triggered.
Enter Cancellation Reason	The Changeset Editor enters the cancellation reason.
Trigger Template Draft Cancellation	Template Draft Cancellation Event is triggered.
Delete Retirement Action	The system deletes any Retirement Actions.
Changeset Cancellation Process Complete Event	Changeset Cancellation Process Complete Event is triggered. <i>*Changeset moves to CANCELED state</i>

Template Correction Subprocess

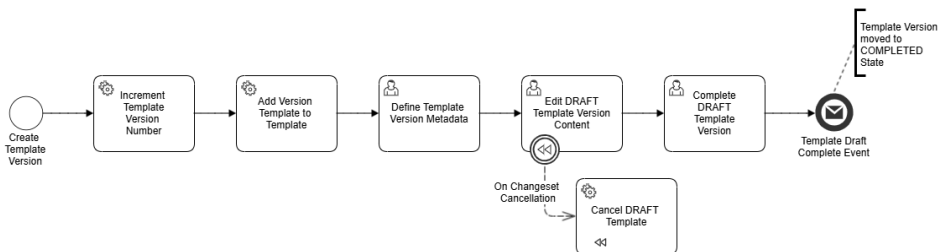
Task	Description
Changeset Test Failed Event	The process begins when a Changeset Test Failed Event is triggered.
Review Test Results	The FAILED test results are reviewed by the Changeset Editor.
Correct Template	The Changeset Editor selects the Template that needs to be corrected.
Template Changes Needed Event	Template Changes Needed Event is triggered.
Make Corrections	The Changeset Editor makes corrections to the Template.
Template Draft Completed Event	Template Draft Completed Event is triggered.
Changeset Draft Completed Event	Once all templates are corrected, a Changeset Draft Completed Event is triggered.

Add Existing Template to Changeset Subprocess (****awaiting final approval**)



Task	Description
Add Existing Template	The process begins with a request to add an existing Template to the Changeset.
Select Add Existing Template to Changeset	The Changeset Editor selects and add an existing Template to the desired Changeset.
Check Template is in "Finished" Changeset	Business rule check to see if the template is in a "Finished" Changeset state. Note: "Finished" Changeset States are Canceled & Published
If Template is in "Unfinished" Changeset Remove Template from Changeset	If the Template is in an "Unfinished" Changeset, the Template will be removed from the Changeset you are trying to add it to, handling the duplicate. A template can only be in one "Unfinished" Changeset at a time.
Existing Template Added to Changeset	The existing Template is added to the Changeset.

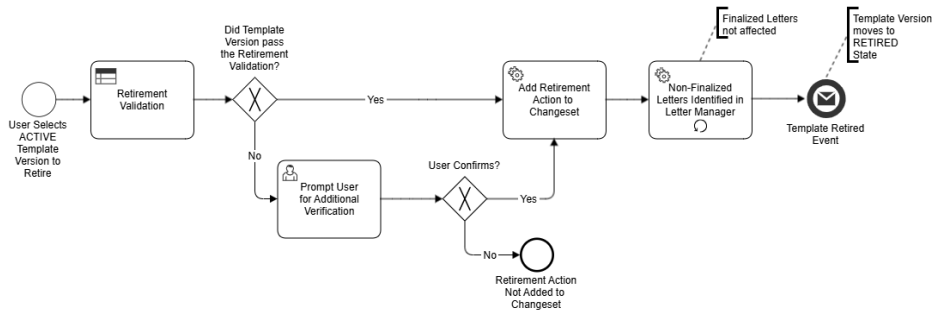
Create Template Version for Changeset Subprocess (**awaiting final approval)



Task	Description
Create Template Version	The process begins with Changeset Editor initiating the creation of a Template Version.
Increment Template Version Number	The system increases the version number for the Template to reflect the new Version.
Add Version Template to Template	The system adds the newly created Template Version to the collection of Templates in the system.
Define Template Version Metadata	The Changeset Editor specifies and records metadata for the new Template.
Edit DRAFT Template Version Content	The Changeset Editor edits and refines the content of the DRAFT Template Version as needed.
On Changeset Cancellation Cancel DRAFT Template	On Changeset Cancellation, the DRAFT Template is CANCELED.
Complete DRAFT Template Version	The Changeset Editor finishes the DRAFT Template Version content, preparing for completion.
Template Draft Complete Event	The Template Draft Completed Event is triggered. <i>*Template Version moves to COMPLETED state</i>

Template Version Retirement Subprocess (**awaiting final approval)

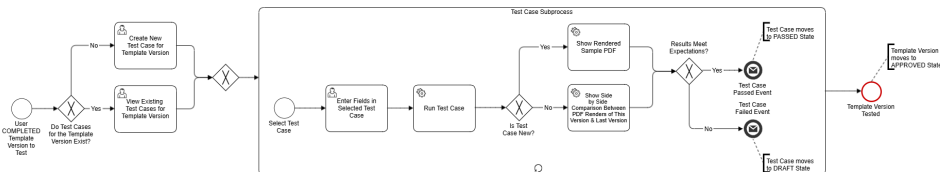
This model details how different Template states are handled during retirement. Template Version retirement is a stand-alone process, but also integral to the Connection Decommissioning process.



Task	Description
User Selects ACTIVE Template Version to Retire	The process starts when the Changeset Editor selects an ACTIVE Template Version to retire.
Retirement Validation	Business rule check to confirm that the Template Version can be retired.
Did Template Version pass the Retirement Validation?	Decision point: Did the template version pass the retirement validation check?
If No Prompt User for Additional Verification	If the Template Version fails the retirement validation, the Changeset Editor is prompted for additional information or verification.
User Confirms?	Decision point: The Changeset Editor is notified of the failed retirement validation and is asked to confirm if they still want to proceed with retiring?
If No Retirement Action Not Added to Changeset	If the Changeset Editor does not confirm that they want to proceed with retiring, the Retirement Action is not added to the Changeset.
If Yes Add Retirement Action to Changeset	If the Template Version passes the retirement validation OR the Changeset Editor confirms, the Retirement Action is added to the Changeset.
Non-Finalized Letters Identified in Letter Manager	All Non-Finalized Letters associated with the Template Version are identified in Letter Manager. Note: Finalized Letters are not affected.
Template Retired Event	Template Retired Event is triggered. <i>*Template Version moves to RETIRED state</i>

Template Version Testing Subprocess (**awaiting final approval)

Before a Changeset is Scheduled and Published we want to make sure the new Template Version(s) reflect the changes we want, so we check that by creating and using Test Cases that let us view Rendered PDF's for those Templates, using sample data. Test Cases are created for specific Template Versions, and are used inside of Changesets to test them, but live outside of the Changeset, and can be re-used by future Changesets to regression test changes to those Templates.

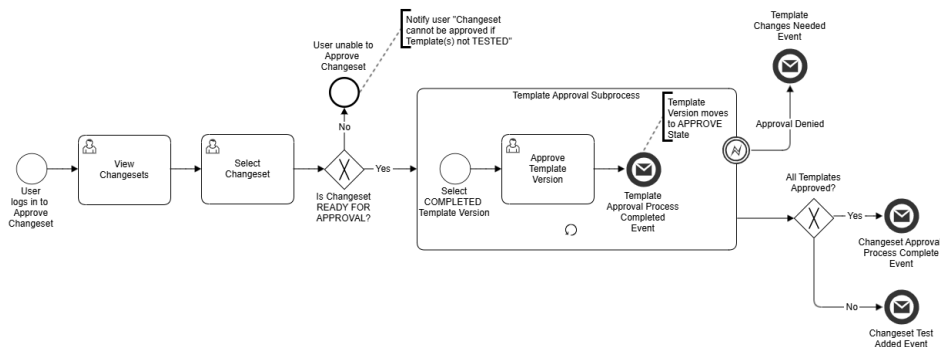


Task	Description
User COMPLETED Template Version to Test	The process starts when the Changeset Editor has a COMPLETED Template Version that they want to test.
Do Test Cases for the Template Version Exist?	Decision Point: determine if Test Case(s) already exist for the Template Version.
If No Create New Test Case for Template Version	If Test Case(s) do not already exist, the Changeset Editor creates a new Test Case for the Template Version.

If Yes View Existing Test Cases for Template Version	If Test Case(s) exist, the Changeset Editor view the existing Test Cases for the Template Version.
Select Test Case	The Changeset Editor selects the appropriate Test Case to work with.
Enter Fields in Selected Test Case	The Changeset Editor enters the required fields for the selected Test Case.
Run Test Case	The system executes the Test Case logic.
Is Test Case New?	Decision point: the system checks if the Test Case is new or not.
If Yes Show Rendered Sample PDF	If the Test Case is new, the system renders and displays the sample PDF.
If No Show Side by Side Comparison Between PDF Renders of This Version and Last Version	If there are existing Test Case(s), the system shows a side-by-side comparison of current and previous PDF Template Versions.
Results Meet Expectations	Decision point: do the results meet the Changeset Editor's expectations?
If Yes Mark Test Case as Passed	If yes, mark the Test Case as PASSED. <i>*Template Version moves to APPROVED state</i>
If No Mark Test Case as Failed	If no, mark the Test Case as FAILED. The Changeset Editor loops back for further action.
Template Version Tested	Template Version Testing is complete.

Changeset Approval Subprocess (**awaiting final approval)

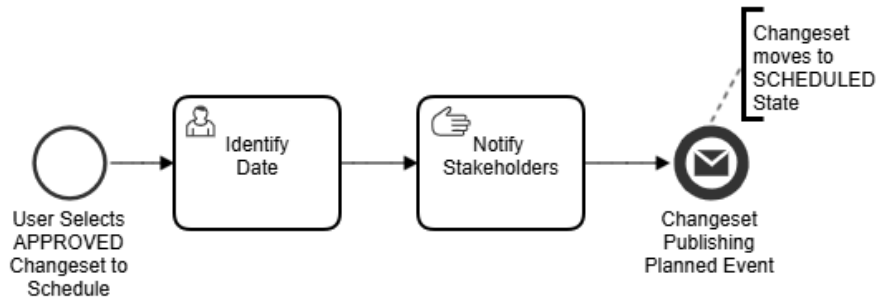
This flow governs the approval of tested Templates within a Changeset. It includes subprocesses for Template approval and cancellation handling.



Task	Description
User logs in to Approve Changeset	The Changeset Publisher logs into Template Manager approve a Changeset.
View Changesets	The Changeset Publisher sees a list of Changeset(s).
Select Changeset	The Changeset Publisher selects the Changeset they want to approve.
Is Changeset READY FOR APPROVAL?	Decision Point: Is the Changeset in "READY FOR APPROVAL" status?
If No User unable to Approve Changeset	If the Changeset is not in "READY FOR APPROVAL" status, the Changeset Publisher is unable to approve the Changeset and are notified: "Changeset cannot be approved if Template(s) not TESTED"
If Yes Template Approval Subprocess	If the Changeset is in "READY FOR APPROVAL" status, the Changeset Publisher will begin the Template Approval Subprocess. This subprocess will be repeated for each template in the Changeset.
Select COMPLETED Template Version	The Changeset Publisher will select a Template Version in "COMPLETED" status.
Approve Template Version	The Changeset Publisher will approve the Template Verison.
Template Approval Process Completed Event	Template Approval Process Completed Event is triggered. <i>*Template Version moves to APPROVED state</i>

Approval Denied Template Changes Needed Event	The Changeset Publisher is able to deny Template(s) as they see fit. If the Template approval is denied, a Template Changes Needed Event is triggered.
All Template Approved?	Decision Point: Are all templates in the Changeset approved?
If No Changeset Test Added Event	If any template(s) were not approved, Changeset Test Added Event is triggered.
If Yes Changeset Approval Process Completed Event	If all template(s) were approved, Changeset Approval Process Completed Event is triggered.

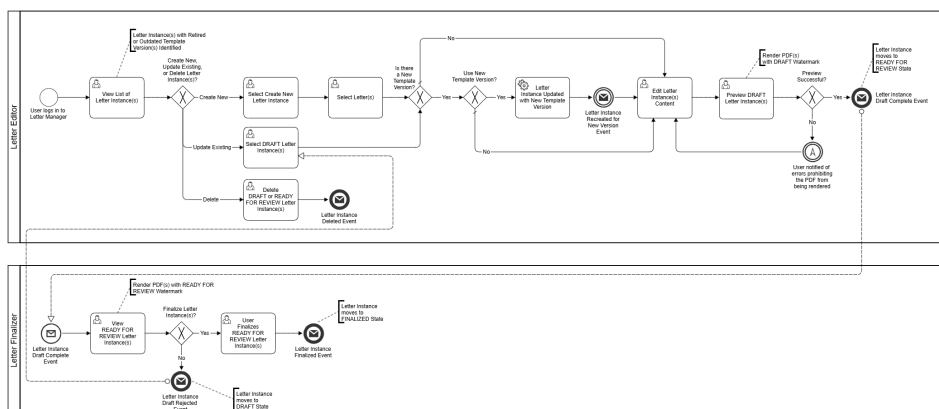
Changeset Scheduling Subprocess (**awaiting final approval)



Task	Description
User Selects APPROVED Changeset to Schedule	The Changeset Publisher selects an APPROVED Changeset to Schedule.
Identify Date	The Changeset Publisher identifies the date/time they want to Schedule the Changeset to be published on.
Notify Stakeholders	The stakeholders are notified of the date/time scheduled.
Changeset Publishing Planned Event	Changeset Publishing Planned Event is triggered. <i>*Changeset moves to SCHEDULED state</i>

Letter Finalization Process

This swimlane model illustrates how Letters are reviewed, approved and Finalized.



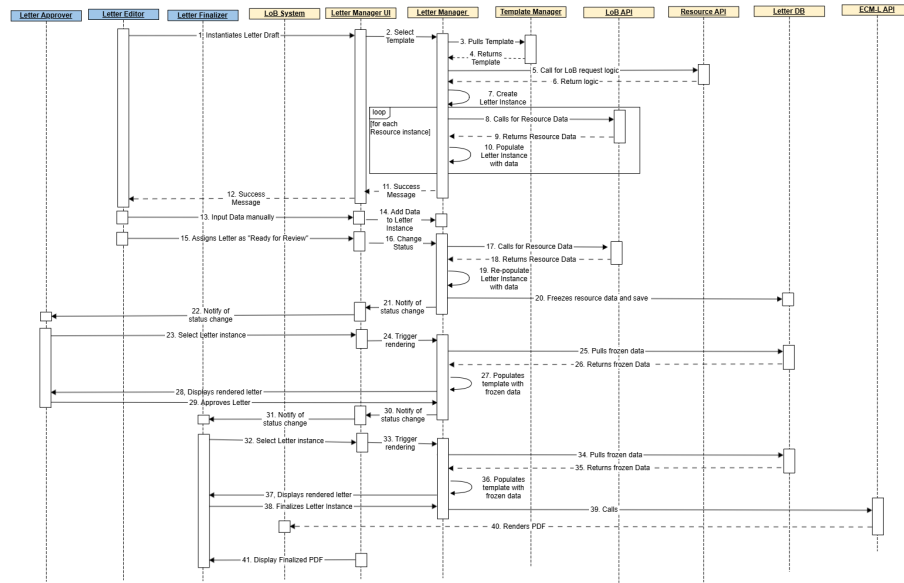
Task	Description
User Logs into Letter Manager	The Letter Editor logs in to Letter Manager.
View List of Letter Instance(s)	The Letter Editor sees a list of existing Letter Instances. Note: Any outdated/retired Templates are identified.

Create New, Update Existing, or Delete Letter Instance(s)?	Decision Point: Do you want to create a new letter, update an existing one, or delete a letter?
If Create New Select Create New Letter Instance	If Create New, the Letter Editor selects create a new Letter Instance.
Select Letter(s)	The Letter Editor chooses which letter(s) to create.
If Update Existing Select DRAFT Letter Instance(s)	If Updating Existing, the Letter Editor selects existing letter(s) in DRAFT status for updating.
If Delete Delete DRAFT or READY for REVIEW Letter Instance(s)	If Delete, the Letter Editor selects letters for deletion triggering a Letter Instance Deleted Event.
Is there a New Template Version?	Decision Point: For New Letter Instances and Update Existing Letters, a check is done to determine if a newer Template Version exists for the letter
If No Edit Letter Instance(s) Content	If there is not a newer Template Version, the Letter Editor will proceed with editing the content of the letter.
If Yes Use New Template Version?	Decision Point: If there is a new Template Version, do you want to use the new Template Version or not?
If No Edit Letter Instance(s) Content	If the Letter Editor does not want to use the new Template Version, they will proceed with editing the content of the letter.
If Yes Letter Instance Updated with New Template Version	Letter Instance Updated with New Template Version.
Letter Instance Recreated for New Version Event	Letter Instance Recreated for New Version Event is triggered.
Edit Letter Instance(s) Content	The Letter Editor will proceed with editing the content of the letter.
Preview DRAFT Letter Instance(s)	The Letter Editor previews the letter as a PDF with DRAFT watermark.
Preview Successful?	Decision Point: Checks if the preview was generated successfully?
If No User notified of errors	The Letter Editor is notified of errors prohibiting the PDF from being rendered.
If Yes Letter Instance Draft Complete Event	Letter Instance Draft Complete Event is triggered. <i>*Letter Instance moves to READY FOR REVIEW state</i>
Letter Instance Draft Completed Event	The Letter Finalizer's process starts once the Letter Instance Draft Completed Event is triggered and the letter instance is in READY FOR REVIEW.
View READY FOR REVIEW Letter Instance(s)	The Letter Finalizer views Letter Instance(s) that are ready for review. PDFs are rendered with READY FOR REVIEW watermark.
Finalize Letter Instance(s)?	The Letter Finalizers chooses whether to Finalize the Letter Instance(s) or not.
If No Letter Instance Draft Rejected Event	If not Finalized, the Letter Instance Draft Rejected Event is triggered. <i>*Letter Instance moves back to DRAFT state</i> (Letter Editor can proceed with updating the letter)
If Yes User Finalizes READY FOR REVIEW Letter Instance(s)	If Finalized, the Letter Finalizer finalizes the Letter Instance(s).
Letter Instance Finalized Event	Letter Instance Finalized Event is triggered. <i>*Letter Instance moves to FINALIZED state</i>

Process Sequence - Systems Event-Trace Description (SV-10c)

The following diagrams illustrate chronological sequences of events and interactions within different workflows of ECM-L, providing a dynamic view of system behavior over time.

Letter Creation Sequence (SV-10c)



Implementation View

The Implementation View (sometimes called the Development View) will be further completed alongside the Development of ECM-L.

Deployment View

Purpose	This section is to document any deviations or unique conditions that are not outlined by the standard practices of the BIP Platform which may be found in the Deployment View section of BIP Architecture Documentation (2001AM).
Acceptance Criteria	- Environments - System Configuration - System Monitoring and Metrics - System Logging and Auditing - Security Strategy
Diagram Type	UML Component or Block if needed

Environment Mapping

Environment	Description
DEV	Development Environment
TEST	Development testing environment
INT	Development Integration testing environment
IV&V/SQA	Independent Verification & Validation / System Quality Assurance
UAT	User Acceptance Testing
PAT	Pre-Acceptance Testing
PROD Development Test (PDT)	Environment used by Development to Test changes and patches
DEMO	Demonstrations
PREPROD	Environment used to test new releases prior to deploying to PROD

PERF	Performance testing
Cost of Living Adjustment (COLA)	Environment used to assist in determining Cost of Living changes
PROD Test (PRODTEST)	Environment to test PROD changes and patches prior to deploying to PROD
PROD	Production

System Configuration

No deltas from BIP standard practices.

System Monitoring and Metrics

No deltas from BIP standard practices.

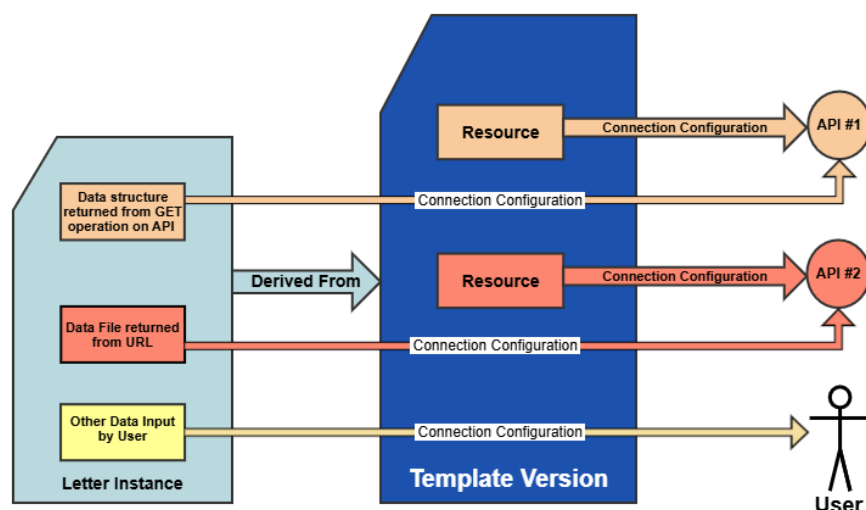
System Logging and Auditing

No deltas from BIP standard practices.

Long-Term Provenance

Letter Instance Provenance refers to the ability to chronicle how a Letter Instance was created, i.e., by whom, and when, while also tracing the origins of every element, including configured content as well as all data values input. A Letter Instance is derived from a Template Version, thus maintaining a comprehensive record of the Template Version, from which it was derived, including the Template Version's unique content and the Resource Version(s) it used, will ensure traceability of all the configured content. Chronicling the data input ensures the verifiable origin of the data necessary for auditing and validation purposes. For all data fields, the record will log the data/time of input, the source of data (e.g., the Resource Version from which the data originated (API or URL) or user, if manually input), and the actual data values entered in the Letter Instance.

The diagram below illustrates how each data element depicted in the Letter Instance (Orange, Red, Yellow) can be traced to its origin. For example, the Data Structure input in the Letter Instance can be traced to the Orange Resource (configured in the Template Version) which used a Connection Configuration to API #1. The Data File can be traced to the Red Resource (configured in the Template Version) which accessed the Data File from API #2 via the Connection Configuration and Other Data can be traced to the User who manually input the data in the Letter Instance.



Security Strategy

ECM-L is heavily dependent on policies concerning authentication and authorization and will utilize the **Benefits Application AuthoriZation** framework (BAAZ) as the propagation mechanism for an LOB's authorization policies. BAAZ is a policy framework that uses Open Policy Agent (OPA) with a central policy store where LOBs can manage their own policies and any application can create its own policies and apply policies that are managed by a central security authority and appropriately audited. (See [Benefit Applications Authorization Services](#)). As an enterprise system, LOBs will have self-service capabilities for maintaining their own governance and security policies. An important part of the Intake Process will be for each LOB to establish a set of policies for that LOB, including policies for access to Applications and Workflows.

Recommendation

Analysis of Alternatives

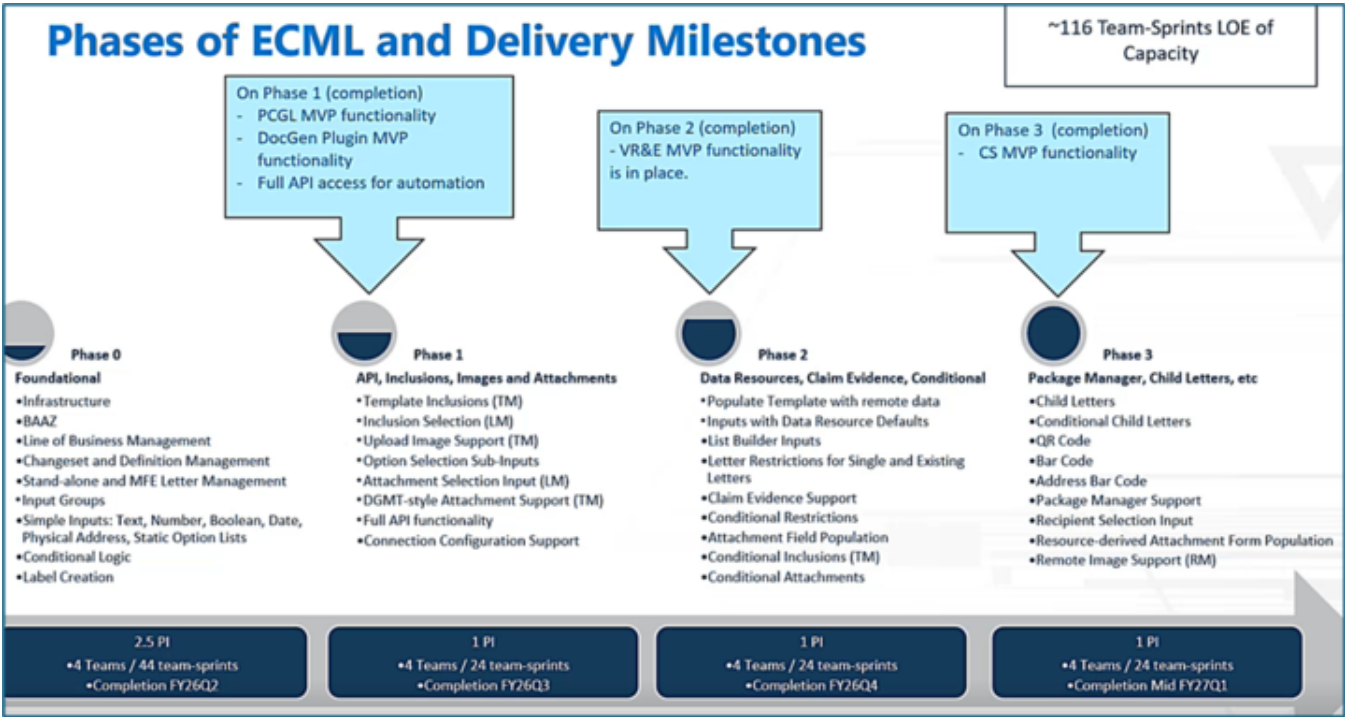
Within the VA, individual Lines of Business (LOBs) use various applications to generate and manage letters, with each application tailored to the specific LOB business needs.

When devising an Analysis of Alternatives (AOA) for a transition to Enterprise Content Manager for Letters (ECM-L), each LOB must consider the cost, risk and benefit of continued use of their current application for letter generation versus implementing ECM-L. Since ECM-L will be implemented in a phased approach, scaling from simple letters to more complex letters, a LOB must also consider the complexity of the letters they need to generate to determine the appropriate transition course. If the complexity of a LOB-generated letter exceeds the capability of ECM-L, an interim solution may be required.

Recommendation

Phases and Roadmap

Phases



Phase Map

Functionality Group	Functionality	Jira Ticket (TBD)
Infrastructure	New Tenant Apps, Repos, etc.	
BAAZ	BAAZ Integration Intake Process	
Line of Business Management	Create Workflow	
	Create Key Parameter Type	
Changeset and Template Management	Create Changeset	
	Create, Update, Retire Template	
	Create and Run Test Cases	
	Schedule Changeset	

	Create, Align Template Label	
	Get Template List	
	Label Creation	
Stand-alone and MFE Letter Management	Create, Edit, Delete Letter	
	Render Letter	
	Finalize Letter	
Input Support	Conditional Logic	
Simple Input Support	Text Inputs	
	Number Inputs	
	Boolean Inputs	
	Date Inputs	
	Physical Address Inputs	
	Option Selection Inputs	
Complex Input Support	Input Groups	
	Attachment Selection Inputs	
	List Builder Inputs	
	Option Selection Sub-Inputs	
	Recipient Selection Input	
	Inputs with Data Resource Defaults	
Inclusions	Template Inclusion	
	Inclusion Selection	
	Conditional Inclusions	
Image Resource Support	Upload Image Support	
Integration Support	Full API Functionality	
Connection Configuration Support	Connection Configuration Support	
Attachment Support	DGMT-Style Attachments	
	Conditional Attachments	
	Attachment Field Population	
	Remote Image Support	
	Resource-Derived Attachment Form Population	
Restrictions	Letter Restrictions for Single and Existing Letters	
	Conditional Restrictions	
Populate Template with Remote Data	Text Fields (Data Resource)	
	Number Fields (Data Resource)	
	Boolean Fields (Data Resource)	
	Date Fields (Data Resource)	

	Physical Address Fields (Data Resource)	
	Field Groups (Data Resource)	
Child Letters	Child Letters	
	Conditional Child Letters	
Bar Code Support	QR Codes	
	Bar Codes	
	Address Bar Codes	
Storage and Distribution	Claim Evidence Support	
	Package Manager Support	